



Data-driven robust optimization scheduling for microgrid day-ahead to intra-day operations based on renewable energy interval prediction

Mao Yang, Jinxin Wang^{*}, Yiming Chen, Yuxuan Zeng, Xin Su

Key Laboratory of Modern Power System Simulation and Control & Renewable Energy Technology, Ministry of Education (Northeast Electric Power University), Jilin, 132012, China

ARTICLE INFO

Keywords:

Microgrid
Renewable energy interval prediction
GPR
Robust optimization
MDQN

ABSTRACT

The uncertainty in renewable energy forecasting significantly impacts microgrid scheduling, and traditional scheduling schemes are often overly conservative and limited by a single time scale, resulting in unreasonable strategies that fail to balance reliability and cost-effectiveness. Based on this, a comprehensive two-stage day-ahead and intra-day microgrid scheduling framework is proposed integrating forecasting, regulation, and decision-making. Based on historical power data of renewable energy, a multi-kernel covariance function is applied to improve Gaussian process regression (GPR) for adaptively generating renewable energy power prediction intervals at various confidence levels and verifies the construction of robust optimization uncertainty scenario sets. In the day-ahead scheduling phase, a two-stage adaptive robust optimization model based on interval probability uncertainty sets is established to ensure minimal scheduling costs under the worst-case scenario. Meanwhile, a modified deep Q network (MDQN) algorithm based on a k-priority sampling strategy is proposed to transform the two-stage iterative process into a discrete Markov decision process for solution. In the intra-day scheduling phase, the day-ahead scheduling scheme is followed, and the intra-day scheduling scheme is optimized through multi-time-scale rolling optimization to reduce the impact of renewable energy power fluctuations. Case studies validate that the proposed scheduling method ensures robustness against uncertainties in renewable energy output while also maintaining the economic efficiency of system operations.

1. Introduction

Microgrid, as effective forms of integrating various types of distributed generation, are considered as viable solutions to the challenges posed by the large-scale integration of renewable energy [1,2]. However, the output of renewable energy is characterized by randomness and volatility, which pose significant challenges to the safe and economic scheduling of microgrid. Therefore, it is crucial to explore how to maximize the utilization of renewable energy while ensuring the safe and economical operation of the system, highlighting the substantial research value and significance of this topic [3,4].

Currently, the main approaches to addressing the uncertainty of renewable energy output in microgrid include spinning reserve capacity [5], stochastic programming [6,7], and robust optimization [8,9]. The main advantage of spinning reserve capacity is its simplicity and effectiveness. While the standby capacity is sufficiently large, it can effectively address uncertainties in the system. However, maintaining standby status for reserve units increases the system's investment costs,

and any unused standby capacity leads to resource wastage, making it economically unviable [10]. Stochastic programming directly uses the probability distributions of random variables, enabling a more accurate depiction of system uncertainties. However, its technical challenge lies in the difficulty of precisely describing the probability distributions of uncertain parameters and scenarios occurrences [11]. Furthermore, as the number of scenarios increases, the complexity of solving the models also rises. The approach of reducing scenarios, which simplifies uncertain variables from their original interval domain to a number of representative scenarios for multi-scenario deterministic optimization, risks losing the traversability of the uncertainty interval, making it difficult to ensure robustness. Moreover, if the clustering algorithm lacks precision, the resulting "typical" scenarios may not possess sufficient representation [12]. Robust optimization, as a powerful tool for addressing uncertainty in stochastic optimization problems, has the advantage of not requiring prior knowledge of the probability distribution of uncertain variables. By describing variable uncertainties in the form of sets, it ensures that constraints are satisfied for all possible

^{*} Corresponding author.

E-mail address: 1436705235@qq.com (J. Wang).

<https://doi.org/10.1016/j.energy.2024.134058>

Received 21 May 2024; Received in revised form 29 October 2024; Accepted 30 November 2024

Available online 2 December 2024

0360-5442/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

values within a known set of uncertain variables. Due to its low requirements for describing uncertain variables, robust optimization has attracted considerable attention in microgrid energy management [13].

The core technique of robust optimization lies in the selection of uncertainty variable sets. He et al. [14] uses box uncertainty variable sets to describe the uncertainty of source-load output, considering the source-load output as fluctuating within a defined interval without considering its specific probability distribution. Qin et al. [15] combines box uncertainty sets with two-stage robust optimization to construct a two-stage robust optimization scheduling model with a min-max-min structure. Yan et al. [16] considers multiple uncertainties in the system, integrating a stochastic hierarchical scenario generation method with a box uncertainty set to address multi-objective optimization and multi-stage coupling problems. In the above references, the construction of the box uncertainty set inadequately considers the adjustment of set conservatism, resulting in limitations in capturing the diversity and dynamic changes of uncertain variables. Consequently, the flexibility of scheduling is constrained, and the decision-making tends to be overly conservative.

To address this drawback, Shi et al. [17] introduces uncertainty constraints into the model to generate polyhedral uncertain sets, thereby adjusting the conservatism of robust optimization results. Wu et al. [18] considers the uncertainties of new energy output and uncoordinated charging of electric vehicles, constructing a robust optimization model with polyhedral uncertainty sets. By capturing the worst-case scenarios under different conditions, the model ensures the system's stable and efficient operation under various unexpected situations. In Ref. [19], a day-ahead to intra-day optimal scheduling framework based on robust optimization and stochastic optimization methods is proposed, employing a polyhedral uncertainty set to represent the uncertainties in wind power and load. Yuan et al. [20] considers the correlations between uncertain variables, employing elliptical uncertainty sets to obtain the wind power time correlation axis and centre point, forming a wind power time correlation polyhedral uncertainty sets. The above studies have improved scheduling flexibility by adjusting the uncertainty of uncertain variables and have reduced the conservativeness of robust optimization to some extent. However, issues remain regarding the overly coarse of uncertain variables and subjectively predetermined target deviation factors. Additionally, insufficient consideration is given to the historical data and distribution information of uncertain variables, which actually disconnects the relationship between prediction and scheduling.

Currently, some scholars have proposed a robust optimization method that uses data-driven representations of system uncertainty sets. Zhang et al. [21] constructs an uncertain set using historical data that combines wind power output and traffic flow, proposing a data-driven two-stage robust scheduling model. Ye et al. [22] develops a wind power uncertain set based on a "closed-loop feedback" data-driven approach and presents a robust scheduling method that accounts for the uncertainty set of historical wind power prediction errors. References [21,22] considers prediction data with varying accuracy during the forecasting stage but insufficiently explores the inherent features of historical data, resulting in suboptimal economic efficiency and robustness of the scheduling scheme in certain extreme scenarios. References [23,24] employs a minimum volume enclosing convex hull uncertainty set, considering the spatiotemporal correlations of wind power, and proposes a data-driven robust optimization model for coordinating the scheduling of multi-energy coupling systems. Zhao et al. [25] uses generative adversarial networks to generate artificial wind power data to improve the estimation of worst-case probability distributions in robust optimization, thereby adjusting the robustness of the optimization solution in terms of cost-effectiveness. The above research models uncertainty through historical data or prediction errors and estimates their distribution using statistical methods. However, it lacks sufficient feature extraction from the historical data and does not incorporate the characteristics contained in the prediction data. Based on

this, references [26,27] constructs a Wasserstein fuzzy set using historical sample data, which accurately captures the prediction errors of source-load output. However, the representational effectiveness of the Wasserstein fuzzy set largely depends on the selected fuzziness parameter and probability level. Inappropriate parameter settings may result in overly conservative scheduling outcomes or inadequate robustness. Additionally, under extreme conditions, the Wasserstein fuzzy set may fail to fully capture actual dynamic changes, potentially limiting the model's ability to effectively address the system's real risks.

Currently, solutions for robust optimization models commonly employ traditional intelligent algorithms or mathematical methods based on robust optimization. Dong et al. [28] proposes a constrained multi-objective transfer algorithm to solve multi-objective optimization problems with the operational costs and robustness of integrated energy systems as optimization goals. Although this method can balance multiple optimization objectives, it has poor convergence and often fails to find the optimal value when dealing with programming problems involving complex constraints. Luan et al. [29] proposes a robust improved particle swarm optimization algorithm, which dynamically determines the optimal solution using the longest distance expectation fitness evaluation strategy in an n -dimensional space. However, due to the high complexity of the algorithm, it can lead to long computation times when handling high-dimensional problems. Shen et al. [30] employs a column constraint generation algorithm to solve a robust optimization model considering renewable energy uncertainty and carbon emissions. However, as the size of the problem increases, the complexity of the model prevents the algorithm from consistently maintaining an optimal solution. Cai et al. [31] uses a distributed algorithm based on the bisection method combined with a column constraint generation algorithm to address the low-level non-convex planning convergence difficulties in two-stage robust optimization models. However, it still faces poor convergence and long computation times when handling complex non-convex problems. Therefore, traditional intelligent algorithms often require a long time for optimization due to cross-iteration and are prone to getting trapped in local extrema. Furthermore, mathematically-based robust optimization methods necessitate the establishment of complex mathematical models, which are less applicable in scenarios with high-dimensional optimization variables.

In addition, the scheduling model constructed in the above studies are concentrated on the day-ahead time horizon. However, as the predictive time-scale shortens, the forecast accuracy of new energy power and load demand improves, thus the day-ahead scheduling plans cannot be directly adopted in the intra-day scheduling. A proper adjustment of the scheduling plans based on more accurate intra-day prediction data is essential. Therefore, it is necessary to study multi-time scale optimization strategies to adapt to different uncertainties caused by prediction accuracy variations. References [32,33] proposed multi-time scale optimization strategies based on the multi-energy flexibility. References [34,35] constructed stochastic optimization and robust optimization models to address uncertainties in new energy output and load demand during day-ahead to intra-day scheduling. However, the consideration of uncertainties in the above studies at the day-ahead stage tends to be overly risky or conservative: the former results in excessive costs for intra-day tracking of the day-ahead plan, sometimes failing to fully mitigate power fluctuations, while the latter may lead to total costs exceeding acceptable levels.

To address the above problems, this paper proposes a two-stage day-ahead and intra-day microgrid scheduling framework integrating forecasting, regulation, and decision-making. The main innovations are as follows.

- 1) Based on historical power data of wind and photovoltaic and combined with the covered probabilistic information, a multi-kernel covariance function is used to improve the GPR algorithm for adaptive generation of prediction intervals. Chance constraints are introduced to combine probabilistic confidence intervals with robust

concepts, applying this approach to construct robust optimization uncertainty sets, thereby achieving correlate between interval prediction and scheduling.

- 2) In the day-ahead scheduling phase, a two-stage adaptive robust optimization model based on interval probability uncertainty sets is established to ensure minimal scheduling costs of microgrid under the worst-case scenario. In the intra-day scheduling phase, the day-ahead scheduling scheme is followed, and the intra-day scheduling scheme is optimized through multi-time-scale rolling optimization to reduce the impact of renewable energy power fluctuations.
- 3) To address the issues of high modelling complexity and low solution efficiency in traditional methods, a k-prioritized sampling strategy is introduced into the DQN algorithm, constructing a MDQN algorithm that transforms the two-stage iterative process into a discrete Markov decision process. The proposed method is validated to effectively reduce model complexity and improve solution efficiency while ensuring the robustness of economic scheduling in microgrid.

2. Based on improved GPR for renewable energy power interval prediction

2.1. Improving GPR model

Gaussian process (GP) is a type of stochastic process capable of effectively handling small-sample and highly stochastic renewable energy power prediction problems. By considering the relationship between the historical power values of wind and photovoltaic as input vector x and their corresponding target values y as $GP(f(x))$:

$$y = f(x) + \varepsilon \quad (1)$$

In the equation, ε represents independent identically distributed noise variables, which follow a Gaussian distribution with a mean of 0 and a variance of σ^2 , denoted as $\varepsilon \sim N(0, \sigma_n^2)$.

The mean and covariance functions of a GP can be represented by the following equations:

$$\begin{cases} m(x) = E(f(x)) \\ k(x, x') = E\{[f(x) - m(x)][f(x') - m(x')]\} \end{cases} \quad (2)$$

In the equation, $x, x' \in R^d$ represents any random variable, d denotes the dimensionality of the input data, and $m(x)$ and $k(x, x')$ are the mean and covariance function.

According to Bayes' theorem, it's necessary to assume a prior distribution for $f(x)$. In GPR, it's common to assume that $f(x)$ follows a Gaussian process with a mean of 0:

$$f(x) \sim GP(0, k(x, x')) \quad (3)$$

Using GPR for renewable energy power interval prediction mainly consists of two stages: model training and testing. In the model training stage: assuming there are n observations in training set $D = \{(x_i, y_i) | i = 1, 2, \dots, n\}$, $X_i = [x_1, x_2, \dots, x_n]$ is an $d \times n$ -dimensional input training matrix and $y_i = [y_1, y_2, \dots, y_n]^T$ is a training output vector composed of n training output scalars y_i . Then, the prior distribution of the observed target values y is:

$$y \sim N(0, K + \sigma_n^2 I) \quad (4)$$

In the equation, $K = K(X, X)$ is a $n \times n$ -dimensional symmetric positive definite covariance matrix, and I is the identity matrix. The output values of n training samples y and the output values of n^* testing samples f^* form a joint Gaussian prior distribution, as follows:

$$\begin{bmatrix} y \\ f^* \end{bmatrix} \sim N\left(0, \begin{pmatrix} K(X, X) + \sigma_n^2 I & K(X, X^*) \\ K(X^*, X) & K(X^*, X^*) \end{pmatrix}\right) \quad (5)$$

In the equation, $K(X, X^*)$ is the $n \times n^*$ -dimensional covariance matrix

between n^* testing output samples and n training output samples, and $K(X^*, X^*)$ is the $n^* \times n^*$ -dimensional covariance matrix of n^* testing output samples.

Traditional GPR still has considerable room for improvement in prediction accuracy. Therefore, this paper adopts a multiple kernel covariance (MKC) function composed of a combination of the quadratic rational and Matern covariance functions for GPR. This establishes the improved Gaussian process regression (IGPR), formulated as follows:

$$C(x, x') = \sum_{i=1}^n K_i(x, x'), i = 1, 2 \quad (6)$$

In the equation, $K_i(x, x')$ represents the i th covariance function.

During the model testing stage: Assuming a new set of input data x^* is given, based on training the GP model with training set (X, y) , the posterior distribution of the highest confidence prediction value y^* for the new input data set x^* is obtained as follows:

$$p(y^* | x^*, X, y) = N(\mu^*, \sigma^*) \quad (7)$$

Where equation (8) represents the predicted mean matrix μ^* , and equation (9) represents the predicted covariance matrix σ^* .

$$\mu^* = K(X^*, X)K(X, X)^{-1}y \quad (8)$$

$$\sigma^* = K(X^*, X^*) - K(X^*, X)K(X, X)^{-1}K(X, X^*) \quad (9)$$

With this, IGPR derives the probability distribution satisfied by the predicted data, obtaining interval predictions under a certain confidence level. Considering that the lower limit of the predicted interval for actual renewable energy power cannot be negative, the predicted confidence interval at confidence level $1 - \alpha$ is as follows:

$$[Lb, Ub] = [\max(0, \mu^* - z_{\alpha/2}\sigma^*), \mu^* + z_{\alpha/2}\sigma^*] \quad (10)$$

In the equation, α represents the significance level; Lb, Ub represents the lower and upper bounds of the confidence interval; $z_{\alpha/2}$ is the two-sided α quantile of the standard normal distribution.

2.2. The representation of uncertain set for renewable energy power prediction based on IGPR

Traditional methods for representing uncertain sets, such as box and interval methods, fail to explore important parameters like the mean and variance of predicted values. This not only disrupts the connection between day-ahead forecasting and scheduling but also results in overly conservative decision outcomes. Therefore, based on the interval power-based uncertain set representation method proposed in Ref. [36], this paper integrates the probabilistic characteristics of renewable energy power into the uncertain set. This approach establishes a link between the forecasting and scheduling stages, enhancing the rationality of scheduling. The constructed uncertain set is shown in equation (11).

$$U_{in} = \begin{cases} \mu^* - z_{\alpha/2}\sigma^* / \sqrt{N} < \mu < \mu^* + z_{\alpha/2}\sigma^* / \sqrt{N} & (\sigma^2 \text{ known}) \\ \mu^* - t_{\alpha}(N-1)s / \sqrt{N} < \mu < \mu^* - t_{\alpha}(N-1)s / \sqrt{N} & (\sigma^2 \text{ unknown}) \\ 0 < \alpha < 1 & 0 < N \leq 30, N \in Z \quad (\text{small sample}) \end{cases} \quad (11)$$

In the equation, μ^* represents the mean of the predicted values; σ^2 represents the variance of the predicted values; N represents the sample size; $z_{\alpha/2}$ represents the double-sided α quantile of the standard normal distribution, achieved by adjusting α to scale the boundaries of the uncertain set; $t_{\alpha}(N-1)$ represents the double-sided α quantile of the t distribution with degrees of freedom $N-1$; s represents the standard deviation of the sample.

The representation of the uncertain set is mechanistically presented

through probabilistic prediction intervals. This setup links the critical information contained in historical data to the uncertain set, minimizing the loss of effective decision information during the prediction stage. It reduces the weight and conservatism of subjective influence factors in the uncertain set, thereby decreasing the modelling error in robust optimization.

This paper employs the following three indicators to assess the accuracy of interval prediction.

- 1) The mean absolute error percentage (MAE%) indicator is selected to measure prediction accuracy. Its specific calculation equation is as follow:

$$MAE\% = \frac{\frac{1}{N} \sum_{i=1}^N |f_i - d_i|}{\frac{1}{N} \sum_{i=1}^N d_i} = \frac{\sum_{i=1}^N |f_i - d_i|}{\sum_{i=1}^N d_i} \quad (12)$$

In the equation, f_i represents the predicted value, and d_i represents the actual value.

- 2) The coverage probability (CP) is employed to evaluate the confidence level of the constructed interval:

$$CP^{(1-\alpha)} = \frac{\beta^{(1-\alpha)}}{N} \times 100\% \quad (13)$$

In the equation, $CP^{(1-\alpha)}$ represents the coverage probability, and $\beta^{(1-\alpha)}$ represents the number of actual values falling within the predicted confidence interval under confidence level $1 - \alpha$.

- 3) The mean width percentage (MWP) of the interval is calculated as follow:

$$MWP^{(1-\alpha)} = \frac{1}{N} \sum_{i=1}^N \frac{U(x_i) - L(x_i)}{y_i} \quad (14)$$

In the equation, $MWP^{(1-\alpha)}$ represents the average width of the interval under confidence level $1 - \alpha$, $U(x_i)$ represents the upper bound of the i -th predicted sample, $L(x_i)$ represents the lower bound of the i -th predicted sample, and y_i represents the true value of the i -th sample.

2.3. Algorithm implementation steps and workflow

To effectively handle the multimodal and uncertainty features of historical data of renewable energy, this paper employs the fuzzy C-means clustering algorithm (FCMCA) to enhance model performance [37]. A multi-kernel covariance function improved GPR algorithm based on a combination of FCMCA's RQ and Matern covariance functions (FCMCA-RQ&Ma-GPR) is developed to achieve interval prediction of new energy power. The specific solution steps and flowchart are as follow.

- 1) Construction of similar-day sample set and data preprocessing: according to the FCMCA algorithm, historical data is classified based on daily types, including different time periods within a day, meteorological factors, seasonal variations, holidays, etc. Similar historical data with characteristics similar to the day to be predicted is then searched for, and equation (15) is used for normalization:

$$\tilde{X}(i) = \frac{X(i) - X_{\min}}{X_{\max} - X_{\min}} \quad (15)$$

In the equation, $\tilde{X}(i)$ represents the normalized data, $X(i)$ represents the original dataset, and $X_{\max}$, $X_{\min}$ represents the maximum and minimum values of the original data.

- 2) Model training: the IGPR model is constructed using a multi-kernel covariance function to capture the relationship between renewable energy power and input features. The model is trained using the training set, adapting to acquire model hyperparameters.
- 3) Model prediction: utilize the trained IGPR model for prediction to obtain the mean and variance of the predicted values, and construct interval prediction results at a certain confidence level (see Fig 1)

3. The robust optimization dispatch of microgrids for day-ahead and intra-day based on interval prediction of renewable energy

The microgrid structure is shown in Fig. 2, including wind turbine (WT), photovoltaic (PV), micro-gas turbine (MT), fuel cell (FC), energy storage system (ESS), and load.

To reduce power fluctuations caused by deviations in renewable energy forecasts for both day-ahead and intra-day periods, a two-stage robust optimization scheduling model for microgrid is established. The model framework is showed in Fig. 3 and representation of rolling optimization process is showed in Fig. 4. The day-ahead scheduling is divided into pre-scheduling and re-scheduling phases. The pre-scheduling phase aims to minimize the startup and shutdown costs of MT and FC by formulating a startup and shutdown plan to address any scenario of concentrated uncertainty. The re-scheduling phase focuses on minimizing the daily generation costs of the microgrid, formulating optimal scheduling plan that ensures minimal day-ahead scheduling costs under the worst-case scenario. The intra-day rolling scheduling follows the day-ahead plan, smoothing power fluctuations of renewable energy through multi-time scale rolling optimization of 1 h and 15 min. The long-term intra-day rolling adjusts MT, FC, and purchased power from the power grid to smooth renewable energy fluctuations, while the short-term rolling adjusts energy storage power and load curtailment to address rapid fluctuations in renewable energy output. At the beginning of each rolling cycle during the intra-day, the forecast values within the prediction horizon are updated, and the output plan within the optimization control horizon is optimized. However, only the planned results for the first time period are issued.

3.1. Day-ahead two-stage robust optimization scheduling Model

- 1) The day-ahead pre-scheduling stage

In this stage, the objective is to minimize the startup and shutdown costs of MT and FC, The objective function for is as follows:

$$\min C_1 = C_{MT}^{on} + C_{MT}^{off} + C_{FC}^{on} + C_{FC}^{off} \quad (16)$$

$$\begin{cases} C_{MT}^{on} = c_{MT}^{on} \sum_{t=2}^{N_T} (u_{MT}^t - u_{MT}^{t-1}), (u_{MT}^t > u_{MT}^{t-1}) \\ C_{MT}^{off} = c_{MT}^{off} \sum_{t=1}^{N_T} (s_{MT}^t - s_{MT}^{t+1}), (u_{MT}^t > u_{MT}^{t+1}) \\ C_{FC}^{on} = c_{FC}^{on} \sum_{t=2}^{N_T} (u_{FC}^t - u_{FC}^{t-1}), (u_{FC}^t > u_{FC}^{t-1}) \\ C_{FC}^{off} = c_{FC}^{off} \sum_{t=1}^{N_T} (s_{FC}^t - s_{FC}^{t+1}), (u_{FC}^t > u_{FC}^{t+1}) \end{cases} \quad (17)$$

In the equation, C_{MT}^{on} , C_{MT}^{off} represent the start-up and shut-down costs of MT; C_{FC}^{on} , C_{FC}^{off} represent the start-up and shut-down costs of FC; c_{MT}^{on} , c_{MT}^{off} are the unit start-up and shut-down costs of MT; c_{FC}^{on} , c_{FC}^{off} are the unit start-up and shut-down costs of FC; u_{MT}^t , s_{MT}^t represent binary variables for MT's on and off states, where $u_{MT}^t = 1$ represents MT in the on state and $s_{MT}^t = 1$ represents MT in the off state; u_{FC}^t , s_{FC}^t represent binary variables for FC's on and off states, where $u_{FC}^t = 1$ represents FC in the on state and $s_{FC}^t = 1$ represents FC in the off state.

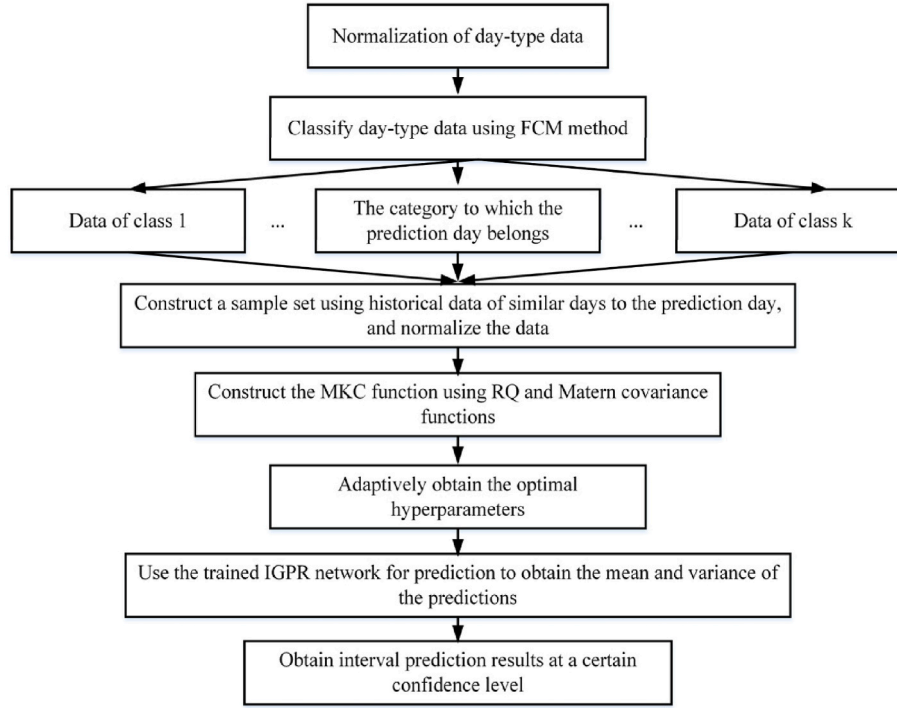


Fig. 1. Algorithm flowchart.

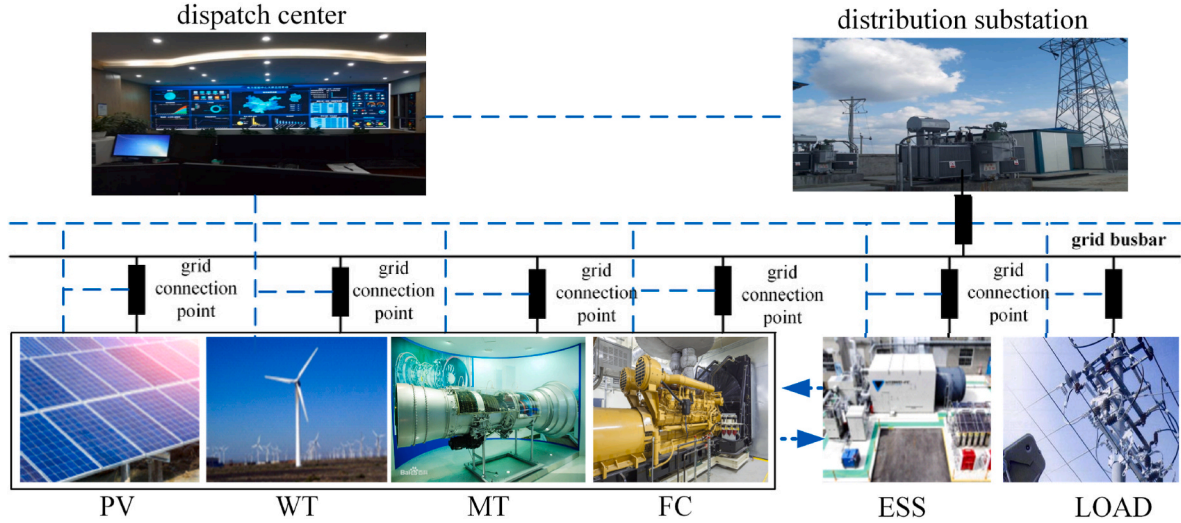


Fig. 2. Microgrid system architecture.

2) The day-ahead re-scheduling stage

Based on the results obtained from the pre-scheduling stage, the day-ahead re-scheduling utilizes IGPR to obtain the day-ahead prediction of renewable energy power. With the operating states of controllable micro-sources determined, a deterministic optimization method is employed to determine the output of devices and the purchase and sale of electricity for the microgrid. The objective function is as follows:

$$\max_{u \in U} \min_{y \in P(x,u)} C_2 = C_{EN} + C_{MT}^e + C_{FC}^e + C_{ESS} + C_{EX} \quad (18)$$

$$C_{MT}^e = \sum_{t=1}^{N_T} k_{MT,OM} P_{MT}^t + c \frac{1}{LHV} \sum_{t=1}^{N_T} \frac{P_{MT}^t}{\eta_{MT}} \quad (20)$$

$$C_{FC}^e = \sum_{t=1}^{N_T} k_{FC,OM} P_{FC}^t + c \frac{1}{LHV} \sum_{t=1}^{N_T} \frac{P_{FC}^t}{\eta_{FC}} \quad (21)$$

$$C_{ESS} = k_{ESS} \sum_{t=1}^{N_T} (P_{dis}^t Y^t - P_{ch}^t X^t) \quad (22)$$

$$C_{EX} = \sum_{t=1}^{N_T} (c_{buy}^t P_{buy}^t - c_{sell}^t P_{sell}^t) \quad (23)$$

In the equation, C_{EN} represents the generation cost of WT and PV, k_{WT} ,

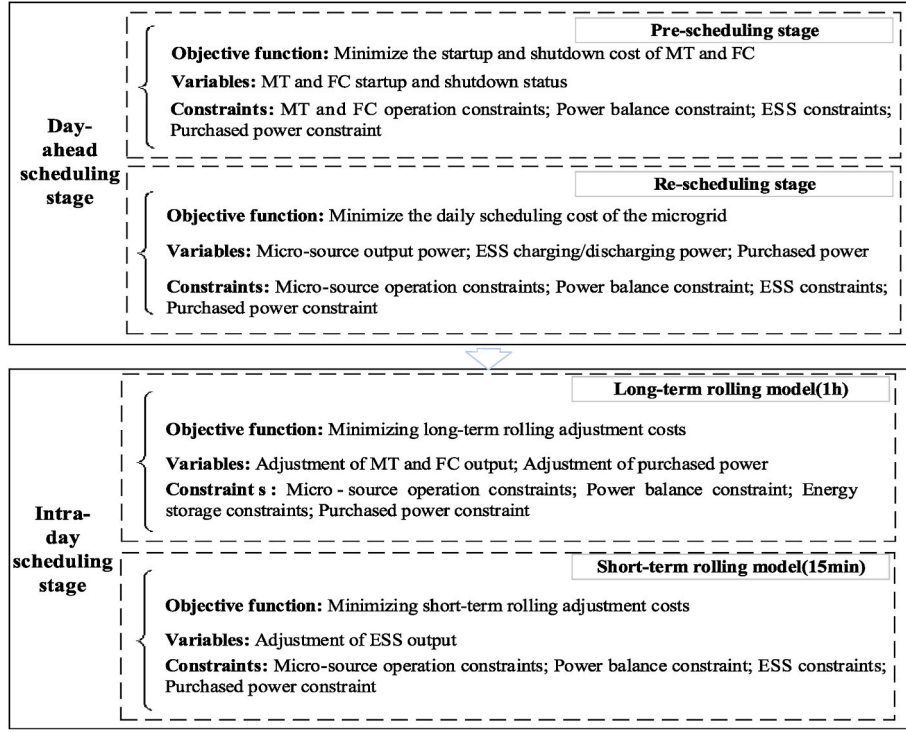


Fig. 3. Robust optimization framework for microgrid in day-ahead and intra-day scheduling.

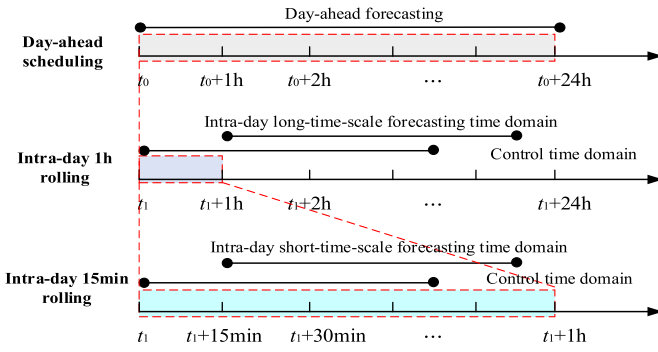


Fig. 4. Representation of rolling optimization process.

k_{PV} represents the unit generation cost of WT and PV, P_{WT}^t, P_{PV}^t represents the day-ahead prediction of WT and PV power for the t -th time period; C_{MT}^t, C_{FC}^t represents the generation cost of MT and FC, $k_{MT,OM}, k_{FC,OM}$ represents the unit operation cost of MT and FC, η_{MT}, η_{FC} represents the generation efficiency of MT and FC, c represents the unit price of natural gas, taken as 2.5 RMB/ m^3 , LHV represents the low heating value of natural gas, taken as 9.7kWh/ m^3 , P_{MT}^t, P_{FC}^t represents the output values of MT and FC for the t -th time period; C_{ESS} represents the generation cost of ESS, k_{ESS} represents the unit operation cost of ESS, X_t and Y_t represent the charging and discharging states of ESS, taking 0 or 1; P_{ch}^t, P_{dis}^t represents the charging and discharging power of ESS; C_{EX} represents the cost of purchasing electricity from the main grid for the microgrid, c_{buy}^t, c_{sell}^t represent the unit purchase and sale electricity prices from the main grid for the t -th time period, P_{buy}^t, P_{sell}^t represent the power of microgrid purchasing and selling electricity to the main grid for the t -th time period.

3) Constraint conditions

MT operation constraints:

$$\begin{cases} 0 \leq u_{MT}^t, s_{MT}^t \leq 1; u_{MT}^t \cdot s_{MT}^t = 0 \\ P_{MT}^{\min} \leq P_{MT}^t \leq P_{MT}^{\max}; |P_{MT}^t - P_{MT}^{t-1}| \leq R_{MT} \end{cases} \quad (24)$$

In the equation, $P_{MT}^{\max}, P_{MT}^{\min}$ represents the upper and lower limits of the MT output, and R_{MT} represents the maximum ramping rate for MT.

FC operation constraints:

$$\begin{cases} 0 \leq u_{FC}^t, s_{FC}^t \leq 1; u_{FC}^t \cdot s_{FC}^t = 0 \\ P_{FC}^{\min} \leq P_{FC}^t \leq P_{FC}^{\max}; |P_{FC}^t - P_{FC}^{t-1}| \leq R_{FC} \end{cases} \quad (25)$$

In the equation, $P_{FC}^{\max}, P_{FC}^{\min}$ represents the upper and lower limits of the FC output, and R_{FC} represents the maximum ramping rate for FC.

Power balance constraint:

$$P_{WT}^t + P_{PV}^t + P_{MT}^t + P_{FC}^t + P_{dis}^t - P_{ch}^t + P_{buy}^t - P_{sell}^t = P_L^t \quad (26)$$

In the equation, P_L^t represents the predicted load power for the t -th time period.

ESS operation constraint:

The formula for calculating the state of charge (SOC) of the ESS is as follows:

$$SOC^t = SOC^0 + \frac{\sum_{t=1}^{N_T} (P_{ch}^t X_t - P_{dis}^t Y_t)}{E_b} \quad (27)$$

In the equation, SOC^0, SOC^t represent the SOC of the ESS for the initial and the t -th time period, while E_b represents the capacity of the ESS. During ESS operation, the following constraints should be satisfied:

$$\begin{cases} 0 \leq X^t, Y^t \leq 1; X^t \cdot Y^t = 0 \\ P_{ch}^{\min} \leq P_{ch}^t \leq P_{ch}^{\max}; P_{dis}^{\min} \leq P_{dis}^t \leq P_{dis}^{\max} \\ SOC^{\min} \leq SOC^t \leq SOC^{\max} \end{cases} \quad (28)$$

To ensure that the ESS has regulation capability at the beginning of the next scheduling cycle, the SOC of the ESS at the start and end of the scheduling period is equal:

$$SOC^0 = SOC^T \quad (29)$$

In the equation, SOC^T represents the SOC of the ESS at the end of the daily scheduling period; $SOC^{\max}$ and $SOC^{\min}$ represent the upper and lower SOC limits of the ESS; $P_{ch}^{\max}$ and $P_{ch}^{\min}$ represent the upper and lower limits of the ESS charging power, and $P_{dis}^{\max}$, $P_{dis}^{\min}$ represent the upper and lower limits of the ESS discharging power.

Microgrid interaction power constraints with the main grid:

$$\begin{cases} 0 \leq P_{buy}^t \leq u_{grid}^t P_{grid}^{\max} \\ 0 \leq P_{sell}^t \leq (1 - u_{grid}^t) P_{grid}^{\max} \end{cases} \quad (30)$$

In the equation, $P_{grid}^{\max}$ represents the interaction power between the microgrid and the main grid, and u_{grid}^t represents the purchase and sale status. When the value of u_{grid}^t is 1, it indicates that the microgrid purchases electricity from the main grid, and when the value of u_{grid}^t is 0, it indicates that the microgrid sells electricity to the main grid.

3.2. Intra-day multi-time scale rolling optimization Model

The intra-day phase adheres to the operational plans derived from the day-ahead scheduling, which includes generation plans for MT and FC, purchasing power plans, and charging/discharging plans for the energy storage system. It smooths power fluctuations of renewable energy through multi-time scale adjustments of 1 h and 15 min.

1) Intraday long-term rolling model

The long-term rolling model is based on the day-ahead plan and aims to mitigate intra-day power fluctuations of wind and photovoltaic over a 1-h time scale by adjusting the outputs of MT and FC, as well as the purchased power from the power grid. The objective function is as follows:

$$\min C_3 = \Delta C_{MT}^e + \Delta C_{FC}^e + \Delta C_{EX} \quad (31)$$

$$\Delta C_{MT}^e = \sum_{t_1=1}^{t_1+T_1-1} \delta_{MT} |\Delta P_{MT}^t| \quad (32)$$

$$\Delta C_{FC}^e = \sum_{t_1=1}^{t_1+T_1-1} \delta_{FC} |\Delta P_{FC}^t| \quad (33)$$

$$\Delta C_{EX} = \sum_{t_1=1}^{t_1+T_1-1} \delta_{EX} |\Delta P_{EX}^t| \quad (34)$$

In the equation, C_3 is the long-term rolling adjustment costs during the intra-day scheduling phase, ΔC_{MT}^e , ΔC_{FC}^e , ΔC_{EX} are the long-term rolling adjustment costs of MT, FC, and purchased power from the power grid, ΔP_{MT}^t , ΔP_{FC}^t , ΔP_{EX}^t are the adjustment power of MT, FC and purchased from the power grid for the t time period, δ_{MT} , δ_{FC} , δ_{EX} are the unit penalty costs for power changes, T_1 is the forecast horizon for the long-term rolling scale.

2) Intra-day short-term rolling model

The short-term rolling scale, based on the long-term scheduling plan, mitigates rapid fluctuations in renewable energy output by adjusting the charging and discharging plan of the ESS. The objective function is as follows:

$$\min C_4 = \sum_{t_1=1}^{t_1+T_2-1} \delta_{SB} |\Delta P_{SB}^t| \quad (35)$$

In the equation, C_4 is the short-term rolling adjustment costs during the intra-day scheduling phase, ΔP_{SB}^t is the adjustment power of ESS for the t time period, δ_{SB} is the unit penalty costs for power changes of ESS, T_2 is the forecast horizon for the long-term rolling scale.

The constraints consider the operational constraints of each generation unit in the day-ahead plan and the power balance constraint, as shown in equations (24)–(30).

4. Model solving

This paper divides the day-ahead scheduling stage into a pre-scheduling and a re-scheduling stage. The constructed two-stage robust optimization model is shown in equation (36). Essentially, it involves solving a min-max-min three-layer optimization problems. The pre-scheduling stage solves the outer min problem, with the decision variable being $\{u_{MT}^t, u_{FC}^t, s_{MT}^t, s_{FC}^t\}$. The optimization variable in this stage ensures that the system can handle any scenario within the uncertainty set. The re-scheduling stage solves the inner max-min problem, with the decision variable being $\{P_{WT}^t, P_{PV}^t, P_{MT}^t, P_{FC}^t, P_{dis}^t, P_{ch}^t, P_{buy}^t, P_{sell}^t\}$. In this stage, after the optimization variable in the pre-scheduling stage is determined, the aim is to find the worst-case scenario and transform the inner model into a deterministic optimization problem. The goal is to optimize the output of each unit under the worst-case scenario while ensuring the minimum day-ahead scheduling cost of the system:

$$\begin{cases} \min_x \{C_1 + \max_{u \in U} \min_{y \in F(x,u)} C_2\} \\ s.t. \text{Equation}(24) - (30) \end{cases} \quad (36)$$

In the equation, C_1 is the objective function of the day-ahead pre-scheduling stage; C_2 is the objective function of the day-ahead re-scheduling stage; U is the uncertainty set of renewable energy output, and u is the uncertainty variable, including WT and PV power; x and y are the decision variables of the pre-scheduling and re-scheduling stages, respectively; $F(x, u)$ is the feasible region formed by the decision variable x in the pre-scheduling stage and the uncertainty variable u .

For the above two-stage robust optimization model, traditional methods decouple it into a master problem and sub-problems, using the column constraint generation (C&CG) algorithm [38,39] to iteratively solve until the upper and lower bounds converge to obtain the optimal value. However, the C&CG algorithm relies on the accuracy and complexity of the model, making it less effective in partially uncertain or dynamically changing environments, and the repeated iterations result in long computation times. In contrast, deep reinforcement learning can gradually learn the optimal behavioral strategy in dynamic and uncertain environments, greatly reducing dependence on complex mathematical models while fully considering various complex situations during training and improving computation speed during solving. Therefore, this paper introduces a k-priority sampling strategy based on the DQN algorithm, constructing a MDQN algorithm capable of solving optimization problems in large-scale discrete action spaces. This algorithm is applied to solve the outer min-max-min problem. After obtaining the optimal solution for the outer problem, the inner model is transformed into a deterministic optimization problem under the worst-case scenario and solved directly using the CPLEX solver. The solving principle of MDQN is as follows.

4.1. Markov decision process

The Markov Decision Process is an effective method for establishing the DQN model [40,41]. It describes how the next state of the system is related not only to the current state but also to the current action strategy. For the outer min-max-min problem addressed in this paper, the Markov Decision Process is constructed, including the state space S , action space A , and reward function R , denoted as $\delta_{MDP} = (S, A, R)$.

- 1) State space design: The state information includes the start-up and shut-down status of MT and FC at each time period. The state space is defined as:

$$s_t = \{t, s_{MT}^t, s_{FC}^t, u_{MT}^t, u_{FC}^t\}, t \propto T \quad (37)$$

- 2) Action space design: The DQN algorithm is a deep reinforcement learning algorithm for discrete actions. For ease of training, two selectable actions, 0 and 1, are set, where 0 represents the unit being shut down and 1 represents the unit being started up. The action space is defined as follow:

$$\mathbf{a}_t = \{s_{MT}^{t+1}, s_{FC}^{t+1}, u_{MT}^{t+1}, u_{FC}^{t+1}\} \quad (38)$$

- 3) Reward function design: The objective function of the pre-scheduling stage is used as the reward term of the reward function, and the power balance constraint, as well as the non-balance constraints such as the operation of MT and FC, are used as the penalty terms of the reward function. To minimize the cost function value of the objective function in the pre-scheduling stage, a negative sign is added in front of the reward function. This way, when the reward function is optimized in the maximum direction, the objective function is optimized in the minimum direction. The reward function is defined as follows:

$$r_t = \begin{cases} -e_f F_{\text{all},t} \beta - e_c C_1 & F_{\text{all},t} > 0 \\ 0 & F_{\text{all},t} = 0 \end{cases} \quad (39)$$

$$F_{\text{all},t} = h^2 + \max(g, 0) \quad (40)$$

In the equation, e_f and e_c are the penalty coefficient for constraints and the cost coefficient for the pre-scheduling stage objective function, respectively; $F_{all,t}$ is the constraint penalty function; h represents the equality constraints of the model; g represents the inequality constraints of the model; and β is the positive transformation coefficient.

4.2. DQN algorithm

DQN algorithm is a reinforcement learning algorithm that combines deep learning and Q-learning. By using experience replay to store the agent's interactions with the environment and randomly sampling a certain amount of these samples for training, it breaks the correlation

between data points, improving the stability and efficiency of the training process. DQN algorithm training process is shown in Fig. 5.

The microgrid optimization model serves as the environment, s_t represents the state information in the environment; a_t is the discrete action corresponding to the maximum value Q chosen by the current value network; r_t and s_{t+1} are the reward and next state obtained after executing the action in the environment, respectively; ω and ω' are the neural network weight parameters of the current value network and the target value network, respectively. To reduce the correlation between samples, the DQN algorithm introduces an experience replay mechanism, where each training step stores $\{s_t, a_t, r_t, s_{t+1}\}$ in the experience replay pool. During training, a certain number of samples are randomly sampled from this pool to update the weights of the neural network, enhancing training stability. Additionally, to prevent instability caused by continuously changing Q -values when training with the same network, the DQN algorithm employs two neural networks: the target value network and the current value network. Initially, the target value network and the current value network have the same structure and parameters. The current value network updates its weight parameters in real-time during training, while the target value network's parameters are frozen. Periodically, the current value network copies its parameters to the target network. Through this process, the DQN algorithm can train the agent stably, enabling it to learn the optimal strategy in the given environment. The DQN algorithm training process mainly includes the following four components:

1) State-action value function

The state-action value function $Q(s_t, a_t; \omega)$ evaluates the action taken in the current state. In the DQN algorithm, the update rule for the state-action value function is as follows:

$$Q_{t+1}(s_t, a_t; \omega) = Q_t(s_t, a; \omega) + \alpha \left[r_t + \gamma \max_{a'} Q'(s_t, a'; \omega') - Q_t(s_t, a; \omega) \right] \quad (41)$$

In the equation, $Q(s_t, a_t; w)$ represents the state-action value function, $Q'(s_{t+1}, a'; w')$ denotes the target value function, α is the learning rate, γ is the discount factor representing the discount of future Q at the current time, and a' denotes the optimal action among all possible actions in state s_{t+1} .

2) Cumulative Reward Function

The cumulative reward function is used in reinforcement learning to evaluate the total reward that can be obtained by executing a certain

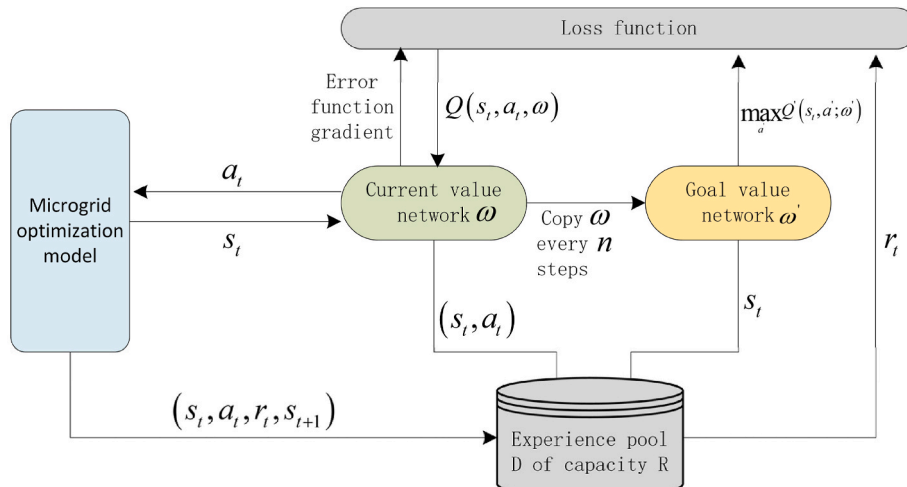


Fig. 5. DQN algorithm training process.

policy starting from a particular state. It represents the sum of cumulative rewards the agent receives from the environment over the long term, guiding the agent to choose better action strategies to maximize the cumulative reward. The calculation of the cumulative reward function is as follow:

$$R_t = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k r_{t+1+k} \quad (42)$$

In the equation, γ is the discount factor, representing the rate at which future rewards are discounted to the present moment.

3) Loss function

The loss function is used to measure the difference between the model's predicted output and the true values. During training, the model adjusts its parameters by minimizing the loss function. The calculation of the loss function is as follow:

$$J(\omega) = Q_t(s_t, a; \omega) + E \left[\left(r_t + \gamma \max_{a'} Q(s_{t+1}, a'; \omega') - Q_t(s_t, a; \omega) \right)^2 \right] \quad (43)$$

4) Action update strategy

The DQN algorithm uses the ϵ -greedy strategy for action selection, where at each action selection, the optimal action is chosen with a probability of $1 - \epsilon$, and other actions are explored with a probability of ϵ . The action update strategy is as follow:

$$a_t = \begin{cases} \arg\max_a Q(s_t, a; \omega), \theta < \epsilon \\ \text{random } a \in A, \theta \geq \epsilon \end{cases} \quad (44)$$

In the equation, $\epsilon \in (0, 1)$ and θ are random variables.

4.3. MDQN algorithm

The ϵ -greedy strategy used in the DQN algorithm helps traverse the action space and balance the exploration rate and exploitation rate in small-scale action spaces. However, in high-dimensional discrete action spaces, the traditional ϵ -greedy strategy is inefficient in exploring non-greedy actions and cannot effectively update the Q network parameters. To address this, this paper introduces a k -priority sampling strategy to replace the ϵ -greedy strategy, constructing a MDQN algorithm. The expression for the k -priority sampling strategy is as follow:

$$\pi(a|s) = \begin{cases} \frac{e^{Q(s_t, a)}}{\sum_{a_k \in A^*} e^{Q(s_t, a_k)}}, a_k \in A^* \\ 0, a_k \notin A^* \end{cases} \quad (45)$$

In the equation, $\pi(a|s)$ represents the policy function, which is a mapping from states to actions.

A^* represents the set of the k actions with the highest action values $Q(s_t, a)$, expressed as:

$$A^* = \{(a_1, a_2, \dots, a_k) | Q(s_t, a_1) \geq Q(s_t, a_2) \geq \dots \geq Q(s_t, a_k) \geq \forall Q(s_t, a_i), a_i \in A\} \quad (46)$$

In the initial phase, the MDQN algorithm is the same as DQN: first, the current state s_t is input into the Q network, and then the action value function $Q(s_t, a_t)$ for all actions is calculated. During the action selection phase, the k -priority sampling strategy as shown in the formula is used. First, k candidate actions with the highest Q values are selected based on the Q values of all actions. Then, the normalized scores of the k candidate actions are calculated using the softmax function. Finally, actions are selected according to the probability distribution that matches the normalized scores.

To summarize, the solution process based on the MDQN algorithm proposed in this paper is shown in Fig. 6.

In the intra-day multi-time scale rolling optimization model, the objective function contains absolute value terms. These can be linearized by introducing auxiliary variables using the method outlined in Ref. [42]. Therefore, the intra-day rolling optimization model can be transformed into a mixed-integer linear model, which can be directly solved using the CPLEX solver. In summary, the solving process for the proposed day-ahead and intra-day two-stage robust optimization model is showed in Fig. 7.

5. Example analysis

To validate the effectiveness of the proposed method in this paper, simulation tests were conducted using the measured WT, PV and load data from a microgrid in Belgium for the entire year of 2018. The simulation parameters are as follows: the power limit for interaction between the microgrid and the power grid is 600 kW, and the parameters of various types of micro-sources in the microgrid are shown in Table 1. The time-based electricity prices of the grid are shown in Table 2. Training parameters of MDQN algorithm in Table 3. Assume that the intra-day rolling phase forecasting time domain T is the same as the control time domain M . The long-term (1h interval) control time domain M_1 is set to 4h, and the short-term (15-min interval) control time domain M_2 is set to 1h.

5.1. Prediction results and error analysis

We utilized the full-year measured data of microgrid in 2018. We used the preceding 270 days' data as the training sets to predict the WT and PV power outputs at 96 time points throughout September 30th. The results of predictions at confidence levels of 95 %, 80 %, and 50 % are depicted in Figures A1 to A2 in appendix. It is evident that both WT and PV predicted expected values effectively track the actual values, exhibiting similar fluctuation trends. Furthermore, as the confidence level decreases, the width of the prediction interval gradually narrows, leading to a decrease in interval coverage rate. This results in more actual power values falling outside the confidence interval, aligning with real-world scenarios and demonstrating the effectiveness of the interval prediction results proposed in this paper.

5.2. Comparison of robust optimization scheduling results

To evaluate the effectiveness of applying the characterization of uncertainty sets to robust optimization. The following three prediction models are set up for comparative verification: Model 1: Using the Matern kernel GPR prediction model(Matern-GPR); Model 2: Using the Matern kernel GPR based on FCMCA(FCMCA-Matern-GPR); Model 3: Using the predictive model proposed in this paper(FCMCA-RQ&Ma-GPR). The interval prediction errors of are shown in Table 4.

1) In the pre-scheduling stage, the unit start-stop status is shown in Table 5. When the confidence level is the same, compared to Model 1, Model 2 adopts FCMCA to classify the historical power data of renewable energy and form a training set for prediction. The MAPE of the predicted expected value decreases, and the MWP decreases when CP is close. The severity of the worst-case scenario that the system needs to deal with decreases. Therefore, the MT and FC startup period decrease; In comparison to Model 2, Model 3 exhibits even fewer operating hours for MT and FC. According to Table 3, Model 3 applies a multi-kernel covariance function to the GPR prediction model, resulting in better interval prediction evaluation indicators compared to a single-kernel covariance function. This improvement is mainly because the WT and PV output sequences contain both linear and nonlinear components. The generalization ability of a multi-kernel covariance function GPR network is better

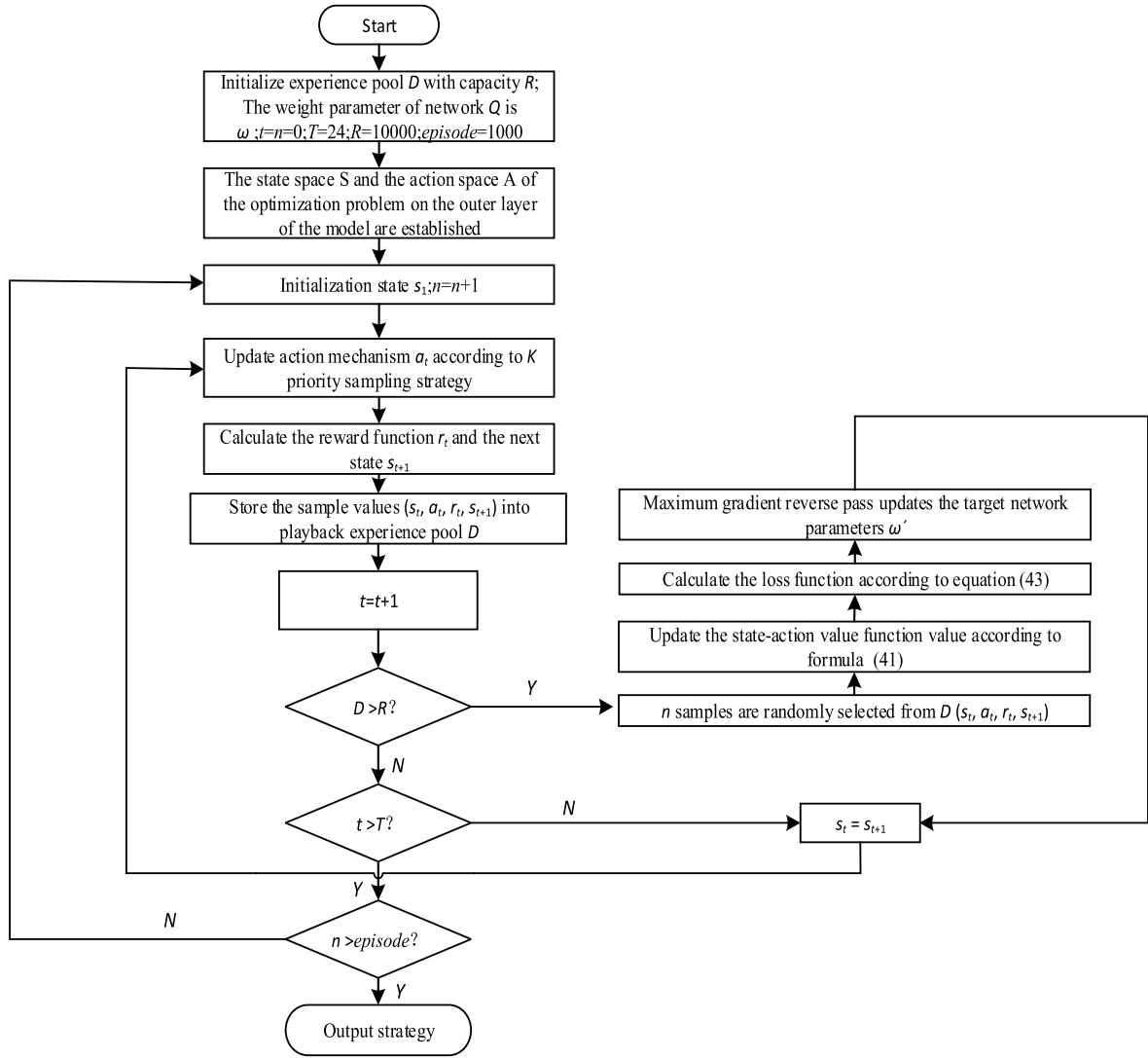


Fig. 6. MDQN algorithm solution process.

than that of a single-kernel function. As a result, the MWP is lower when CP is close, leading to a further reduction in the severity of the worst-case scenario and requiring even less spinning reserve. Therefore, Model 3 has fewer operating hours for MT and FC compared to Model 2. For the same model, as the confidence level increases, the operating hours of MT and FC also increase. This is mainly due to the increase in the MWP of the uncertainty sets constructed by each model with the increase in confidence level. The wider MWP requires the system to address more severe worst-case scenarios, necessitating more spinning reserve, and thus increasing the operating hours of MT and FC.

- 2) In the rescheduling stage, the scheduling plans for the three models are depicted in Figs. 8–10. With the increase in the severity of the worst-case scenario that the system needs to address for WT and PV output as the confidence level rises, the amount of reserve capacity invested during the pre-scheduling stage increases. Consequently, the overall output power of MT and FC increases. At the same confidence level, Model 3 constructs uncertainty sets with narrower intervals for WT and PV output uncertainty, particularly when CP is close. Compared to Models 1 and 2, this allows Model 3 to represent the severity of the worst-case scenario that the system needs to address with lower severity. Consequently, Model 3 requires the least amount of reserve capacity and overall output from MT and FC, making its scheduling plan more effective.

- 3) Combining Table 6 and Figs. 8–10, the standby capacity invested in the pre-scheduling stage of MT and FC, as well as the overall output in the rescheduling stage, both increase. This leads to an over-reservation of standby capacity, consequently driving up the total cost of the microgrid. However, at the same confidence level, Model 3 demonstrates the best economic performance. This is attributed to the superior generalization ability of Model 3's network compared to Models 1 and 2, resulting in better performance across various interval prediction evaluation indicators. Particularly, its MWP is lower when CP is close, requiring less standby capacity reservation by the system. Consequently, Model 3 exhibits lower MT and FC generation costs, thus yielding superior economic benefits.
- 4) Robustness Analysis: Calculations from Table 6 reveal that when the confidence level increases from 50 % to 95 %, the total operating costs of Model 1, Model 2, and Model 3 increase by 8.9 %, 6.8 %, and 4.9 % respectively. Model 3 exhibits the lowest rate of increase in total operating cost, indicating that with the increase in uncertainty disturbance, the variation of the objective function value of the system using Model 3 is the smallest. This reflects the lower sensitivity of the objective function to the fluctuation of WT and PV in the system's optimization scheduling. Therefore, Model 3 demonstrates stronger system robustness in its scheduling results.

The above analysis verifies that applying the prediction model

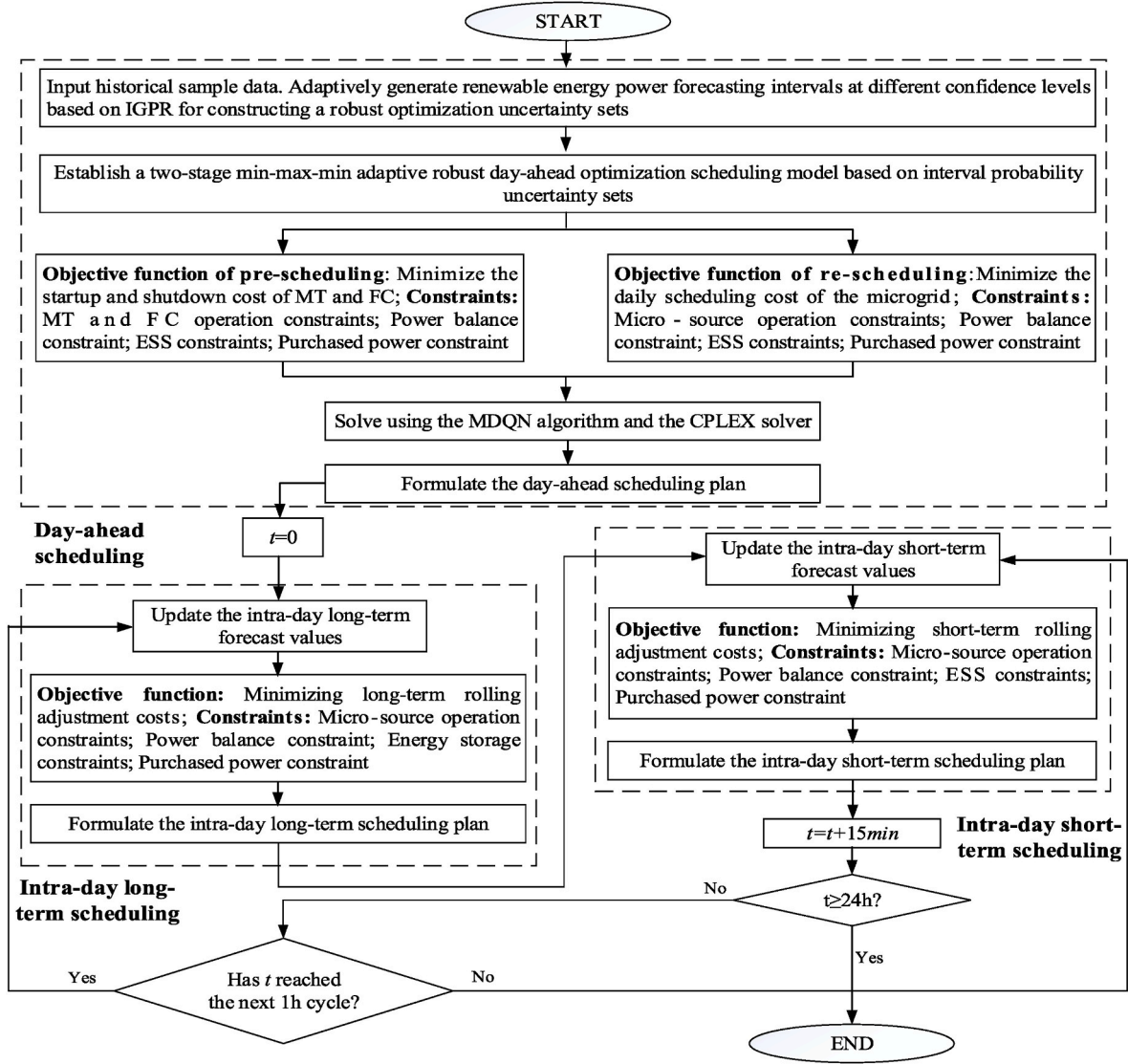


Fig. 7. The model solving process.

proposed in this paper to construct the uncertainty set of renewable energy output for dispatching achieves better results. The dispatch plan formulated by this model can better cope with the uncertainty of renewable energy output, reducing the conservatism of dispatch decisions while ensuring the economic efficiency and robustness of system operation.

5.3. Comparison and validation of uncertainty set representation methods

To validate the effectiveness of the proposed method for representing new energy uncertainties based on interval probability uncertainty sets in robust optimization, the following methods are compared: Method 1 uses the adaptive robust optimization approach with μ -type uncertainty sets [43]; Method 2 uses the adaptive robust optimization approach with Γ -type uncertainty sets [44]; Method 3 uses the adaptive robust optimization approach with uncertainty sets generated by the Dirichlet process [45]; Method 4 uses the adaptive robust optimization approach

with the novel uncertainty sets proposed in this paper. The composition of microgrid operating costs for different methods is shown in Table 7.

From Tables 7 and it can be observed that, under the accuracy of the predicted data for each method, the operating costs for all four methods increase with the confidence level. This is because, at higher levels of conservatism, the uncertainty set boundaries become more extensive, leading to larger intervals of new energy power that need to be considered, thereby increasing operating costs. Under the same confidence level, Method 4 has the lowest operating cost. This is mainly because the uncertainty set constructed by Method 4 captures the characteristics of the new energy output prediction data sequence and incorporates these characteristics into the robust optimization scheduling process, significantly extracting the probabilistic distribution information of the new energy output. Although Method 3 can adaptively cluster sub-uncertainty sets based on new energy output prediction data and better capture the distribution of the prediction data through the union of multiple sub-uncertainty sets, its convergence performance in

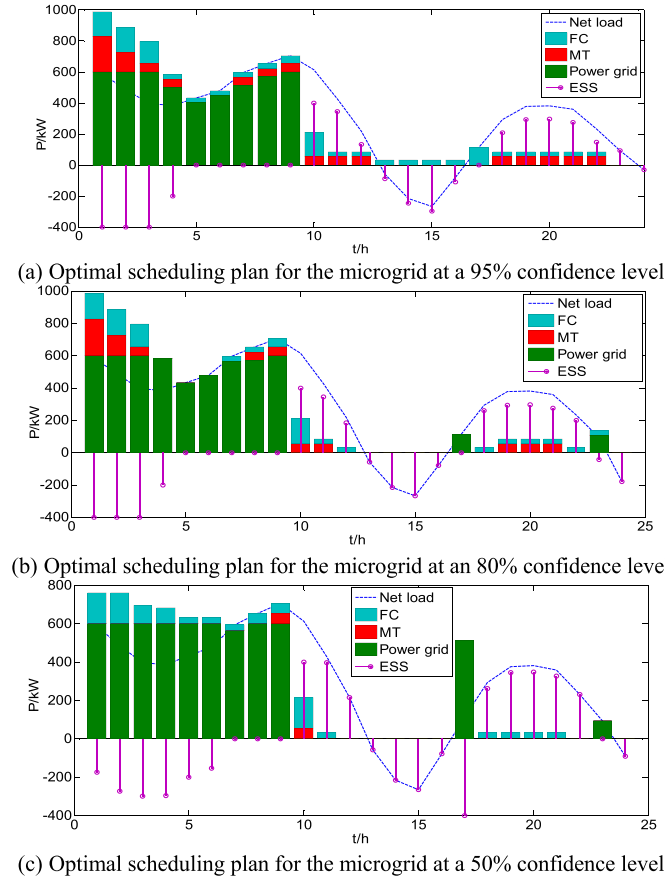


Fig. 8. Optimal scheduling plan for Model 1 at different confidence levels.

clustering sub-uncertainty set boundaries declines during periods of frequent jumps in new energy output values. The compactness of the union boundary of its sub-uncertainty sets is inferior to that of Method 4's interval probability uncertainty set, resulting in slightly higher total operating costs for Method 3 compared to Method 4. Methods 1 and 2, while using prediction data with higher accuracy than Methods 3 and 4, mechanically apply different precision prediction data available in the prediction phase without incorporating the features of the prediction data sequence into the scheduling process. As a result, the uncertainty sets constructed by Methods 1 and 2 cannot adaptively adjust the boundaries of each set according to the characteristics of the prediction data, resulting in overly broad boundaries. Therefore, the operating costs for both Methods 1 and 2 are higher than those for Methods 3 and 4. In summary, compared to Methods 1, 2, and 3, the microgrid economics under the scheduling results of Method 4 are superior, validating the effectiveness of applying the interval probability uncertainty set to robust optimization.

5.4. Comparison and validation of optimization algorithm

In a personal computer configured with an Intel(R) Core(TM) i7-8750H CPU @ 2.20 GHz 2.21 GHz, the MDQN algorithm and DQN algorithm were used to solve Model 3, and the results are shown in Fig. 11. The trend in reward values from the figure indicates that the MDQN algorithm has better convergence characteristics than the DQN

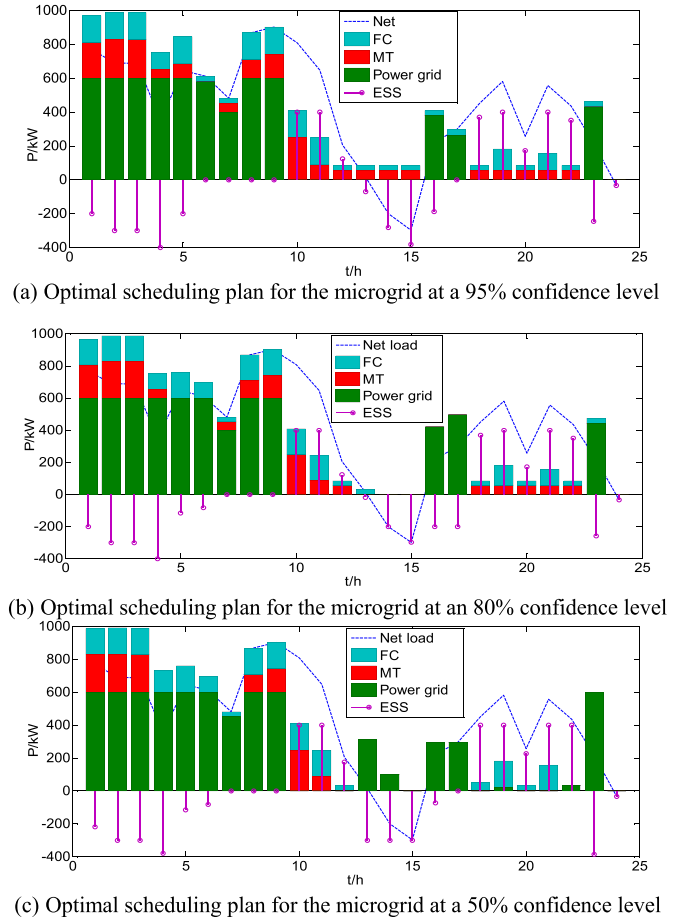


Fig. 9. Optimal scheduling plan for Model 2 at different confidence levels.

algorithm, achieving superior converged reward values and exhibiting smaller fluctuations in reward values during the training process. The main reason is that the DQN algorithm uses the ϵ greedy strategy, which is inefficient for exploration in large action spaces. It fails to learn the optimal strategy within a limited time, and multiple random explorations lead to significant fluctuations in reward values during the learning process, resulting in poorer algorithm robustness. In contrast, the MDQN algorithm proposed in this paper uses the k priority sampling strategy, which selects better candidate actions based on probability. This greatly enhances the agent's exploration efficiency in the action space and ensures that the algorithm converges to better reward values. At the same time, the k priority sampling strategy avoids excessive visits to low-reward actions, thereby reducing the variance of the reward function during the learning process. Therefore, the MDQN algorithm demonstrates better convergence and stability compared to the traditional DQN algorithm. In this case study, the MDQN algorithm achieved a mean reward of 2242.6, while the DQN algorithm achieved a mean reward of 1993.6. The reward variance for the MDQN algorithm was 4.16, compared to 10.24 for the DQN algorithm. Compared to the DQN algorithm, the MDQN algorithm increased the reward value by 11.1 % and reduced the variance by 59.3 %. Thus, the proposed MDQN algorithm significantly outperforms the traditional DQN algorithm in large-scale action spaces.

To further validate the effectiveness of the proposed algorithm, the

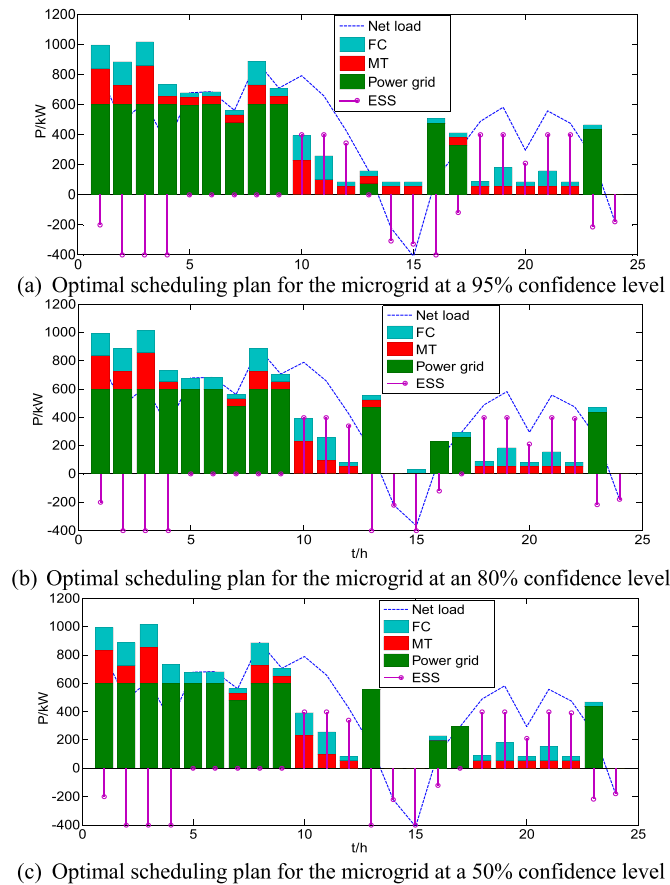


Fig. 10. Optimal scheduling plan for Model 3 at different confidence levels.

Table 1

Parameters of various types of micro-sources.

Sources	Parameters	Values
WT	Unit operation and maintenance cost(RMB/kWh)	0.045
PV	Unit operation and maintenance cost(RMB/kWh)	0.0096
MT	Unit operation and maintenance cost(RMB/kWh)	0.0776
	Unit fuel cost(RMB/kWh)	0.5
	Rated power(kW)	260
	Output upper and lower limits(kW)	[52,260]
FC	Unit operation and maintenance cost(RMB/kWh)	0.0293
	Unit fuel cost(RMB/kWh)	0.4
	Rated power(kW)	160
	Output upper and lower limits(kW)	[32,160]
ESS	Unit operation and maintenance cost(RMB/kWh)	0.045
	Rated capacity(kWh)	2000
	State of charge upper and lower limits	[0.1,0.9]
	Charging and discharging power threshold limits(kW)	[-400,400]

Table 2

Time-based electricity prices of the grid(RMB/kWh).

Categories	Time periods	Purchase electricity price	Selling electricity price
Off-peak period	0–6h	0.17	0.14
Standard period	7–9h; 13–17h; 22–23h	0.54	0.49
Peak period	10–12h; 18–21h	0.83	0.68

Table 3

Training parameters of MDQN algorithm.

Parameters	values	Parameters	values
Training iterations	5000	Replay buffer size	10000
Optimization step size	24	Batch size	64
Target network update frequency	1000	Discount factor	0.9
Learning rate	0.0001	Number of neural network layers	3
Number of neurons in hidden layers	128	Regularization coefficient	0.001

Table 4

Interval prediction errors for the three models.

Confidence level	Indicators	Model 1		Model 2		Model 3	
		WT	PV	WT	PV	WT	PV
95 %	MAE%	7.8	4.6	6.5	3.3	6.0	3.1
	CP%	92	100	100	100	100	100
	MWP%	0.611	0.43	0.52	0.35	0.41	0.30
80 %	MAE%	7.8	4.6	6.5	3.3	6.0	3.1
	CP%	77.2	85.5	75.5	90.5	79.9	92.6
	MWP%	0.55	0.39	0.32	0.27	0.21	0.21
50 %	MAE%	7.8	4.6	6.5	3.3	6.0	3.1
	CP%	50.5	76	55	76.3	55.2	73.5
	MWP%	0.44	0.32	0.20	0.20	0.11	0.14

DQN algorithm, column constraint generation (C&CG) algorithm, genetic algorithm (GA), and MDQN algorithm were used to solve the data from September 21 to September 30 over a period of 10 days. The average calculation results are shown in Table 8.

As indicated in the table, the MDQN algorithm achieves a reduction in computation time and better cost values compared to the DQN algorithm, demonstrating the superior convergence and stability of the improved algorithm. Although the C&CG algorithm has high accuracy and its cost function values are nearly the same as those of the MDQN algorithm, it relies on complex mathematical models and has longer computation times. The GA, on the other hand, involves extensive iterative search for the optimal solution through repeated crossover, resulting in substantial computation time and a tendency to get trapped in local optima, leading to excessively high operating costs. Therefore, the comparison validates the effectiveness of the proposed algorithm.

Moreover, in terms of GPU resource utilization, the MDQN algorithm achieves a GPU occupancy rate of 88 % during training, effectively leveraging the parallel computing capabilities of the GPU to accelerate the training of the deep neural network. In contrast, the DQN algorithm has a GPU occupancy rate of 72 %, indicating a reduced utilization efficiency compared to the MDQN algorithm. Both the GA and C&CG exhibit low GPU resource utilization, limiting their ability to handle complex problems. In terms of CPU resource utilization, the DQN algorithm has a CPU occupancy rate of 48 %, reflecting a relatively high CPU burden. The MDQN algorithm employs a prioritized sampling strategy, which allows it to probabilistically select better candidate actions, significantly enhancing the exploration efficiency within the action space, and its CPU occupancy rate is 26 %, indicating an effective allocation of computational tasks and a reduced CPU burden. In comparison, the GA has a CPU occupancy rate of 86 %, while the C&CG algorithm has a CPU occupancy rate of 77 %, highlighting the traditional algorithms' relatively poor performance in computational efficiency and resource management. Overall analysis reveals that the MDQN algorithm not only demonstrates exceptional GPU utilization but also maintains a relatively low CPU occupancy rate, showcasing its efficiency and adaptability in handling complex tasks.

Table 5

Unit start-stop status in pre-scheduling stage.

Models	Model 1						Model 2						Model 3					
	95 %		80 %		50 %		95 %		80 %		50 %		95 %		80 %		50 %	
	MT	FC	MT	FC	MT	FC	MT	FC	MT	FC	MT	FC	MT	FC	MT	FC	MT	FC
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1
2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1
3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1
4	1	1	1	1	0	1	1	1	1	1	0	1	1	1	0	0	0	1
5	1	1	0	1	0	1	1	1	0	1	0	1	0	1	0	0	0	1
6	1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	0	0	1
7	1	1	1	1	1	1	1	1	1	1	0	1	1	1	0	1	0	1
8	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1
9	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
10	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
11	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1
12	1	1	1	1	1	1	1	1	1	1	0	1	1	1	0	1	0	0
13	1	1	1	1	0	0	1	1	0	1	0	0	0	1	0	0	0	0
14	1	1	0	1	0	1	1	1	0	0	0	0	0	1	0	0	0	0
15	1	1	0	1	0	0	1	1	0	0	0	0	0	1	0	0	0	0
16	0	1	0	0	0	1	0	1	0	0	0	0	0	1	0	0	0	0
17	1	1	0	1	0	0	0	1	0	0	0	0	0	1	0	0	0	0
18	1	1	1	1	1	1	1	1	1	1	0	1	1	1	0	1	0	1
19	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	0	1
20	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	0	1
21	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	0	1
22	1	1	1	1	1	1	1	1	1	1	0	0	1	1	0	1	0	0
23	0	1	0	1	0	1	0	1	0	1	0	0	0	0	0	1	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 6

Day-ahead operating costs of microgrid under different models (RMB).

Model	Confidence level	WT and PV operation cost	Interconnection cost with the main grid	Energy storage operation cost	MT cost	FC cost	Total cost
1	95 %	1812.8	2074.2	235.77	1170.7	910.8	6204.3
	80 %	1812.8	2117.1	244.6	920.1	865.4	5960
	50 %	1812.8	2209.6	234.8	720.6	720.8	5698.6
2	95 %	1856.7	1914.6	235.3	1096.6	887.3	5990.5
	80 %	1856.7	2058.6	235	855.6	789.1	5795
	50 %	1856.7	2096.5	231.2	708.7	715.2	5608.3
3	95 %	1994.4	1357.7	196.3	593.2	558.8	4700.4
	80 %	1994.4	1564.8	202.1	443.2	329.6	4534.1
	50 %	1994.4	1671.2	226.4	260.6	327.8	4480.4

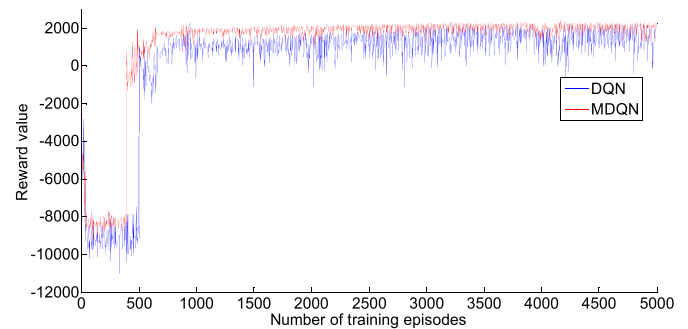
Table 7

Composition of microgrid operating costs for different methods (RMB).

methods	MAE%		Confidence level	Total operating cost
	WT	PV		
1	2.7	1.8	95 %	5011.3
			80 %	4901.3
			50 %	4833.5
2	3.4	2.9	95 %	5112.3
			80 %	4935.5
			50 %	4763.5
3	6.9	4.4	95 %	4821.6
			80 %	4726.2
			50 %	4600.8
4	6.0	3.1	95 %	4700.4
			80 %	4534.1
			50 %	4480.4

5.5. Intra-day multi-time scale rolling optimization scheduling analysis

The scheduling results obtained from Model 3 are used as the day-ahead scheduling plan, and the power modification results for each generation unit during the intra-day long-term rolling phase are shown in Figs. 12–14. It can be observed that as the confidence level increases, the purchased power during the intra-day long-term modification phase becomes higher and more volatile. This is primarily because, when

**Fig. 11.** Comparison of training convergence between MDQN algorithm and DQN algorithm.

formulating the day-ahead scheduling plan, a higher confidence level results in greater MT and FC reserve allocated during the pre-scheduling phase, leading to more flexible power adjustments during the intra-day modification phase. Conversely, a lower confidence level means less system reserve capacity, necessitating compensation for the reserved backup generation through purchasing power, resulting in higher purchased power and greater power fluctuations. The power modification results for ESS during the intra-day short-term rolling phase are shown in Fig. 15, demonstrating that ESS can effectively reduce the magnitude

Table 8
Results of different algorithms.

Algorithm	Solution time/s	Day-ahead operating costs/RMB	GPU occupancy rate (%)	CPU occupancy rate (%)
MDQN	32.6	5120.2	88	26
DQN	85.7	5216.3	72	48
C&CG	112.3	5129.5	19	77
GA	169.5	5546.9	12	86

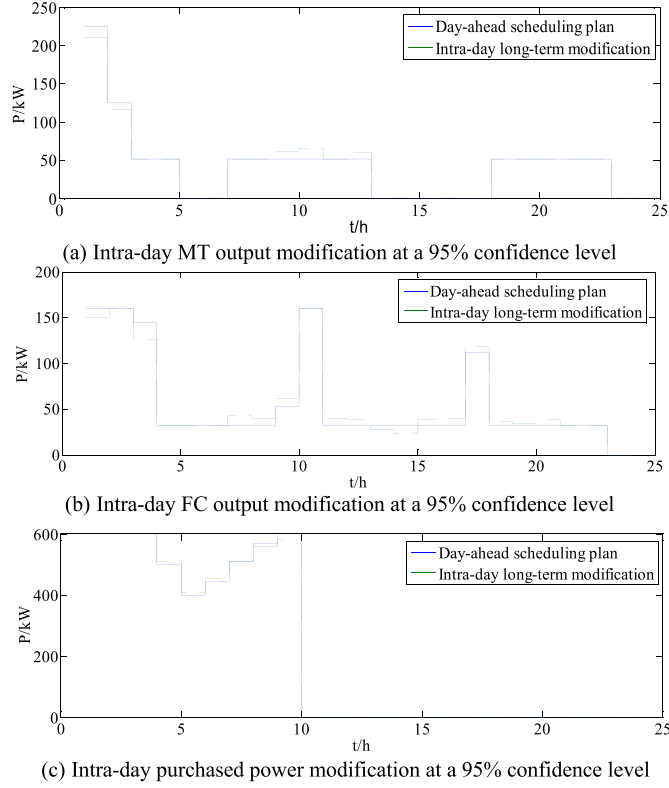


Fig. 12. Intra-day long-term modification at a 95 % confidence level.

of rapid power fluctuations during the intra-day short-term, with good tracking performance across different confidence levels.

To further illustrate the advantages of the intra-day rolling modification phase under the multi-time scale optimization strategy, this study compares the proposed method with the DA-P (day-ahead programming) intra-day power modification strategy at a 95 % confidence level. The DA-P strategy refers to the situation where power imbalances caused by intra-day and day-ahead forecast errors are solely smoothed by the power grid based on the day-ahead scheduling plan. The costs of the two intra-day modification strategies and their interactions with the upper-level grid's power fluctuations are showed in Table 9.

From Tables 3 and it can be seen that the multi-time scale rolling strategy takes into account the fluctuation characteristics of different energy sources. The long time scale mitigates power fluctuations by adjusting the output of MT and FC units, while the short time scale further reduces power fluctuations rapidly through ESS. Because the time scales are hierarchically refined and contribute to smoothing power fluctuations, the strategy maintains a lower level of purchased power fluctuations while improving the consumption rate of renewable energy.

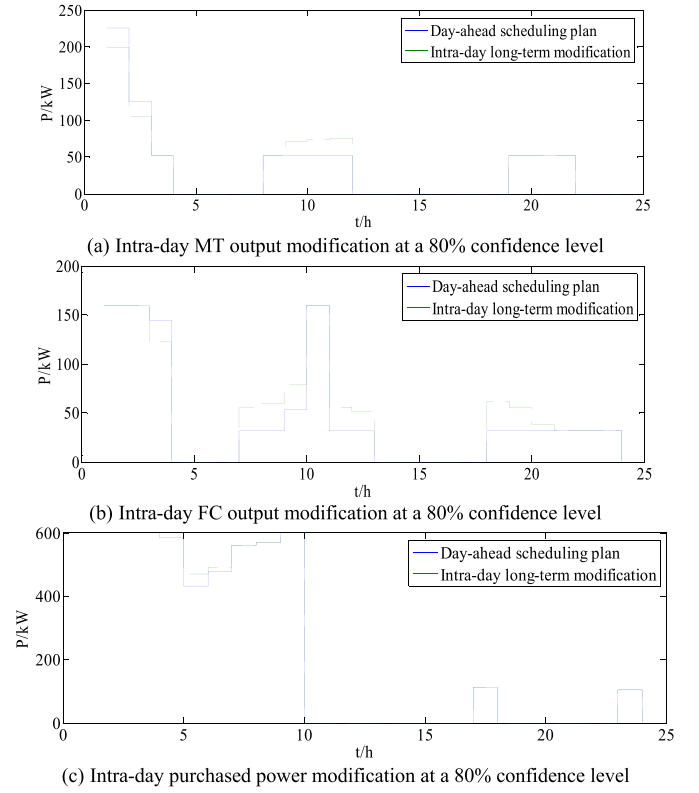


Fig. 13. Intra-day long-term modification at a 80 % confidence level.

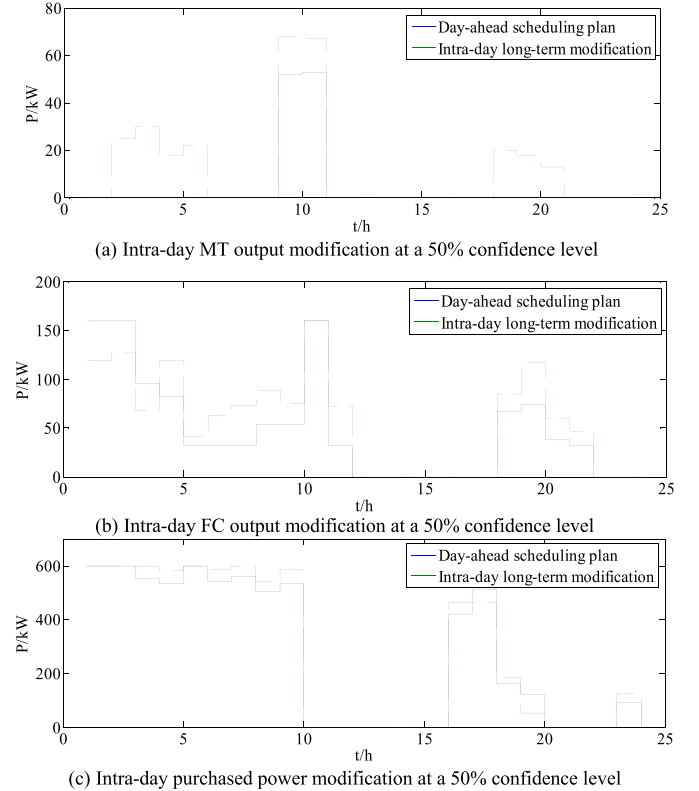


Fig. 14. Intra-day long-term modification at a 50 % confidence level.

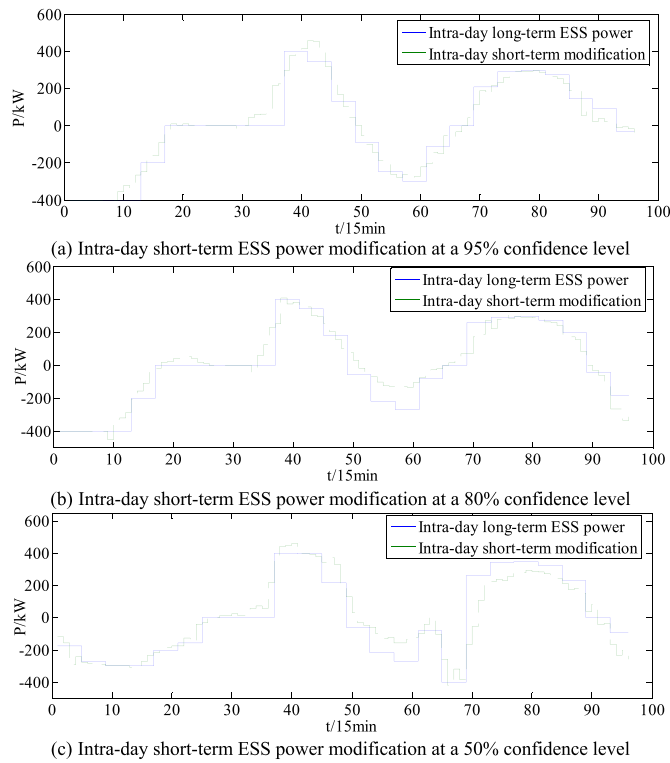


Fig. 15. Intra-day short-term ESS power modification at different confidence levels.

Table 9
Intra-day operating costs of microgrid under different models.

Methods	Renewable energy consumption rate(%)	Total costs (RMB)	Electricity purchase volatility (%)
The method proposed in this paper	96.7	5599.6	7.5
DA-P	83.2	6012.5	33.2

Table 10
Different weather types corresponding to dates.

Categories	Sunny day	Windy weather	Cloudy weather
1	April 6	March 10	May 6
2	April 11	April 15	June 15
3	April 20	April 22	September 28
4	September 11	September 5	October 26
5	October 15	September 26	November 8

Table 11
Sunny day-calculation results.

Methods	Average renewable energy consumption rate (%)	Average day-ahead operating cost (RMB)	Average intra-day operating cost(RMB)	Average actual costs (RMB)
1	82.3 %	5437.5	698.3	6135.8
2	84.3 %	5498.6	740.2	6238.8
3	94.8 %	5473.6	651.7	6125.3
4	94.5 %	5400.3	720.4	6120.7

Table 12
Windy weather-calculation results.

Methods	Average renewable energy consumption rate (%)	Average day-ahead operating cost (RMB)	Average intra-day operating cost(RMB)	Average actual costs (RMB)
1	70.6 %	4726.5	591.2	5317.7
2	82.2 %	4513.6	551.3	5064.9
3	92.2 %	4506.3	465.2	4971.5
4	87.6 %	4287.4	525.4	4812.8

Table 13
Cloudy weather-calculation results.

Methods	Average renewable energy consumption rate (%)	Average day-ahead operating cost (RMB)	Average intra-day operating cost(RMB)	Average actual costs (RMB)
1	89.9 %	6836.4	937.5	7773.9
2	92.6 %	6617.6	951.5	7569.1
3	96.5 %	6512.3	728.5	7240.8
4	95.3 %	6231.5	882.6	7114.1

The DA-P strategy only smooths power fluctuations through purchased power, which leads to excessive power fluctuations on the interconnection lines and a decrease in renewable energy consumption. Therefore, compared to the DA-P strategy, the proposed multi-time scale rolling strategy results in a 25.7 % reduction in the system's purchased power fluctuation rate, a 13.5 % increase in renewable energy consumption rate, and a 6.8 % decrease in total costs.

For the intra-day phase's solving time, the model proposed in this paper was simulated on a personal computer configured with an Intel(R) Core(TM) i7-8750H CPU @ 2.20 GHz 2.21 GHz. The average solving time for a single intra-day short-term rolling optimization is only 0.51s. This is because the intra-day phase is based on multi-time scale theory, significantly shortening the time window length for single optimization and reducing the granularity of the optimization. Consequently, the time required for a single short time scale rolling optimization is also correspondingly reduced. The method proposed in this paper can effectively and in real-time issue scheduling instructions and control for generation devices and ESS within the microgrid during the intra-day phase.

5.6. Model comparison and validation under extreme weather scenarios

To verify the effectiveness of the proposed model in the paper in handling extreme weather scenarios, data from five days each of sunny, windy, and cloudy weather types were selected for comparison and validation. The dates for each weather type are shown in Table 10. The following four models are compared:

Model 1: Data-driven polyhedral uncertainty set optimization model [19];

Model 2: Data-driven Dirichlet process-generated uncertainty set robust optimization model;

Model 3: Data-driven Wasserstein uncertainty set robust optimization model [26,27];

Model 4: The model proposed in this paper.

The average calculation results of the four models under different weather types are shown in Tables 11–13.

The average daily operating cost of Model 1 under sunny weather conditions is close to that of the proposed model in the paper, but under windy and cloudy weather conditions, the average daily operating cost is higher than that of the proposed model in the paper. The main reason

is that when constructing the polyhedral uncertainty set, the upper and lower bounds of the uncertainty set are determined based on the existing forecast accuracy and error, making it difficult to ensure that the polyhedron can cover most possible error scenarios. Consequently, the uncertainty set cannot adaptively adjust the boundaries of each interval based on the characteristics of the forecast data, and its boundaries include a substantial amount of invalid renewable energy fluctuation regions. Especially under extreme scenarios, the day-ahead scheduling plan formulated is overly conservative, resulting in excessively high day-ahead scheduling costs and high actual costs. Additionally, Model 2's limited responsiveness to changes in extreme weather scenarios in a lower renewable energy consumption rate.

Model 2 has a higher average daily operating cost under all three weather types compared to the proposed model in the paper. As previously analyzed, the uncertainty set generated by the Dirichlet process clusters the forecast data corresponding to each time period of the day ahead, incorporating the data characteristics implied by the clustering results into the construction of the sub-uncertainty set boundaries. In contrast, the improved GPR used in this paper incorporates the characteristics of the forecast data (mean and variance) captured at each moment point by point into the construction of the sub-uncertainty set boundaries. This results in the uncertainty set boundaries containing more detailed and precise information about the forecast data. Therefore, under all three weather types, the proposed model in the paper has lower actual costs and higher renewable energy consumption rate compared to Model 2.

Model 3, based on the Wasserstein uncertainty set, addresses output fluctuations caused by extreme weather by constructing a larger uncertainty set. It focuses more on solutions for the 'worst-case' scenario, thus pre-allocating sufficient scheduling resources for the system in extreme weather conditions. While this approach enhances the robustness of the system during extreme weather and allows for more flexible intra-day adjustments, effectively mitigating the additional costs associated with intra-day scheduling, the excessively high costs of the day-ahead plan still result in actual operating costs that are higher than those of the proposed model in the paper. However, due to the broader nature of the uncertainty set constructed by Wasserstein, the actual renewable energy consumption rate is higher than that of the proposed model in the paper.

In summary, by comparing the actual operating costs and renewable energy consumption rate of each model under extreme weather scenarios, it is demonstrated that the proposed model in the paper effectively reduces actual operating costs while improving new energy utilization efficiency. This highlights its superiority and adaptability in variable environments, enabling it to better address the complexities and uncertainties of different extreme weather scenarios. Furthermore, it further validates its practicality and reliability in practical application.

6. Conclusions

This paper proposes a robust optimization scheduling method for day-ahead and intra-day microgrid that integrates prediction, adjustment, and decision-making. The method uses an improved GPR algorithm to adaptively generate the interval prediction of renewable energy output under different confidence levels, linking it to the construction of the robust optimization uncertainty set. A MDQN algorithm is introduced to transform the robust model into a discrete Markov decision process for solution. Through case analysis, the following conclusions

are drawn.

- 1) By comparing the interval prediction performance of different prediction models, it is verified that the proposed FCMCA-RQ&Ma-GPR prediction method can focus more uncertainty information within narrower interval boundaries. The constructed uncertainty set has a stronger overall probability coverage ability at the same confidence level, proving the effectiveness of the proposed prediction model.
- 2) The proposed adaptive robust optimization method based on interval probability uncertainty sets is compared with adaptive robust optimization methods using μ type uncertainty sets, Γ type uncertainty sets, and Dirichlet process-generated uncertainty sets. The results show that the proposed method has the lowest operating cost for the microgrid at different confidence levels. The scheduling plan formulated by this method better copes with the uncertainty of renewable energy output, reducing the conservatism of scheduling decisions while ensuring the economic efficiency and robustness of system operation.
- 3) By comparing the proposed MDQN algorithm with the DQN algorithm, C&CG algorithm, and GA, it is shown that the proposed algorithm can effectively reduce model complexity and improve solving efficiency while ensuring the optimal cost function value of the objective function, demonstrating the superior solving performance of the MDQN algorithm.

The application premise of the proposed adaptive robust optimization model based on interval probability uncertainty sets requires a large amount of historical new energy power data. Insufficient data may lead to inadequate mining of the probability information and characteristics contained in the historical new energy power data. Additionally, this paper's exploration of extreme weather scenarios remains insufficient, which may result in the final decision being overly "risky." Next, extreme weather scenarios will be considered, and incremental learning algorithms will be used to characterize system uncertainty, applying them to the robust optimization model to further improve the accuracy of microgrid scheduling decisions under uncertainty, avoiding overly "conservative" or "risky" decisions to the greatest extent possible.

CRedit authorship contribution statement

Mao Yang: Writing – review & editing, Funding acquisition, Formal analysis. **Jinxin Wang:** Writing – original draft, Methodology, Data curation, Conceptualization. **Yiming Chen:** Resources, Project administration. **Yuxuan Zeng:** Validation, Software. **Xin Su:** Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was supported by the Industrial Technology R&D Program of Jilin Province (Collaborative optimization and operation of low-carbon smart building integrated energy system, 2023C033-5).

Appendix A

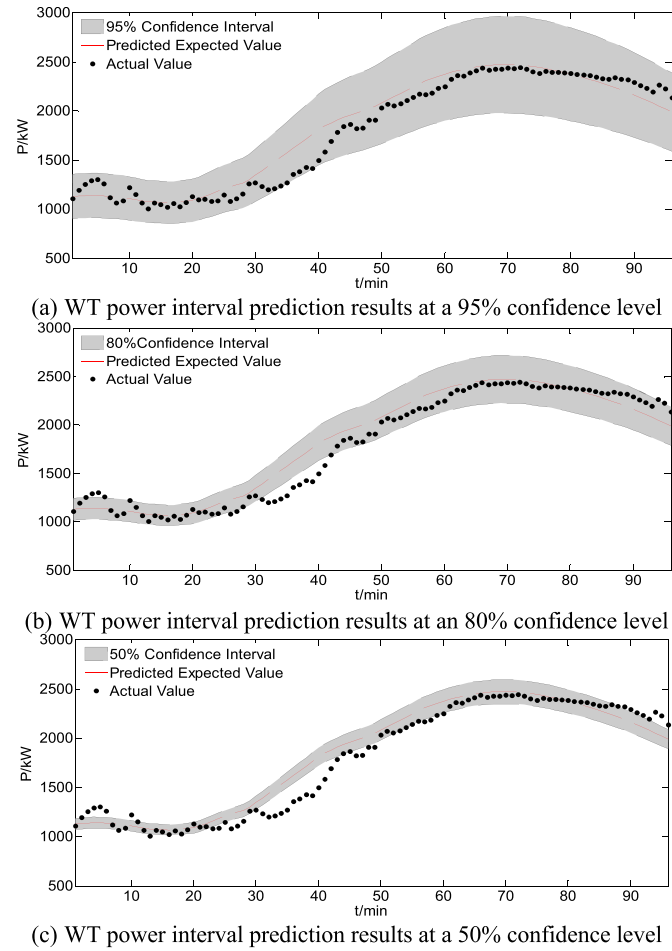


Fig. A.1. WT interval prediction results at different confidence levels.

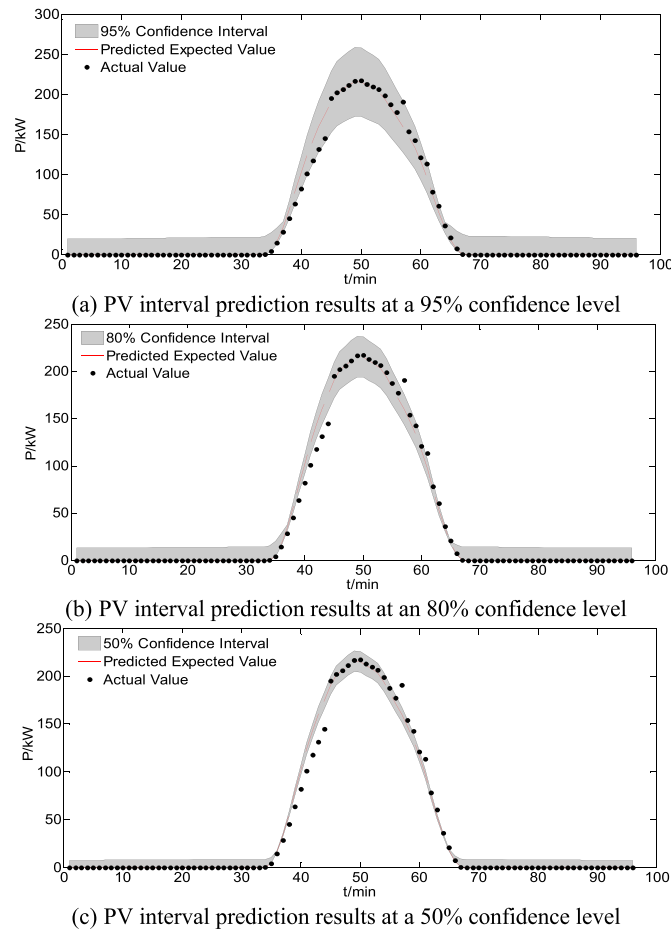


Fig. A.2. PV interval prediction results at different confidence levels.

Data availability

The data that has been used is confidential.

References

- [1] Shi M, Chen X, Shahidepour M, et al. Observer-based resilient integrated distributed control against cyberattacks on sensors and actuators in islanded AC microgrids. *IEEE Trans Smart Grid* 2021;12(3):1953–63.
- [2] Espina E, Cárdenas-Dobson R, Simpson-Porco J, et al. A consensus-based distributed secondary control optimization strategy for hybrid microgrids. *IEEE Trans Smart Grid* 2023;14(6):4242–55.
- [3] Li Y, Ren X, Ge L, et al. Study on full life-cycle planning and operation of electrolytic-hydrogen coupled microgrid based on reversible solid oxide fuel cells [J/OL]. In: *Proceedings of the Chinese society for electrical engineering*; 2024-03-25. p. 1–16.
- [4] Zhao B, Li D, Wu Z, et al. Capacity optimization configuration of island microgrid group based on 100% green energy supply target. *Proc Chin Soc Electr Eng* 2021; 41(3):932–46.
- [5] Wen Y, Lin X. Evaluation of minimum inertia requirement for microgrids under islanded and grid-connected modes. *Proc Chin Soc Electr Eng* 2021;41(6):2040–53.
- [6] Wen X, Chen X, Zhang A, et al. Risk control method for stochastic scheduling market of wind-solar complementary microgrid with adjustable loads. *Power Syst Technol* 2021;45(11):4308–18.
- [7] Liu J, Luo Y, Hou Y, et al. Research on optimization algorithm for active energy management of microgrid considering generalized energy storage. *Power Syst Technol* 2023;47(1):245–55.
- [8] Liu Z, Wang L, Ma L. A transactive energy framework for coordinated energy management of networked microgrids with distributionally robust optimization. *IEEE Trans Power Syst* 2020;35(1):395–404.
- [9] Qiu H, Gu W, Xu Y, et al. Tri-level mixed-integer optimization for two-stage microgrid dispatch with multi-uncertainties. *IEEE Trans Power Syst* 2020;35(5): 3636–47.
- [10] Cui Y, Guan Y, Li J, et al. Low-carbon economic dispatch of source-load-storage considering coordinated operation of carbon capture units and hydrogen storage systems[J/OL]. *Power Syst Technol* 2024;1–13. -03-25].
- [11] Lu M, Li X, Li F, et al. Two-stage stochastic planning for seasonal hydrogen storage - hydrogen blend gas turbine system. *Proc Chin Soc Electr Eng* 2023;43(18): 6978–92.
- [12] Sang B, Zhang T, Liu Y, et al. Two-stage robust optimization scheduling of grid-connected microgrid under expected scenarios. *Proc Chin Soc Electr Eng* 2020;40 (19):6161–73.
- [13] Zhang W, Wang W, Wang H, et al. Capacity optimization configuration of wind-solar-rich oxygen coal-fired power generation system considering carbon capture technology. *Autom Electr Power Syst* 2023;47(13):176–89.
- [14] He C, Wu L, Liu T, et al. Co-optimization scheduling of interdependent power and gas systems with electricity and gas uncertainties. *Energy* 2018;159:1003–15.
- [15] Qin J, Pan X, Sun X, et al. Impact of duplicate charges on two-stage robust optimization scheduling of integrated energy systems and joint trading of carbon-green certificates[J/OL]. *Electric Power Automation Equipment*. 2024.
- [16] Yan R, Wang J, Wang J, et al. A two-stage stochastic-robust optimization for a hybrid renewable energy CCHP system considering multiple scenario-interval uncertainties. *Energy* 2022;247:123498.
- [17] Shi J, Bao Y, Chen Z, et al. Robust optimization configuration method for energy storage of charging stations considering uncertainty of charging load. *Autom Electr Power Syst* 2021;45(20):49–58.
- [18] Wu G, Yi C, Xiao H, et al. Multi-objective optimization of integrated energy systems considering renewable energy uncertainty and electric vehicles. *IEEE Trans Smart Grid* 2023;14(6):4322–32.
- [19] Ji Y, Xu Q, Zhao J, Yang Y, Sun L. Day-ahead and intra-day optimization for energy and reserve scheduling under wind uncertainty and generation outages. *Elec Power Syst Res* 2021;195:107133.
- [20] Yuan S, He Y, Dai C, et al. Polyhedral uncertainty modeling and robust optimization in unit commitment considering wind temporal correlation. *Acta Energiae Solaris Sin* 2020;41(9):293–301.
- [21] Zhang Y, Zheng F, Shu S, et al. Data-driven robust optimization scheduling for electric-gas-transportation coupled systems considering multiple uncertainties. *Proc Chin Soc Electr Eng* 2021;41(13):4450–62.

- [22] Ye L, Zhang B, Guo K, et al. Two-stage cooperative optimization scheduling of wind-fire joint power generation system from day-ahead to intra-day[J/OL]. In: Proceedings of the Chinese Society for electrical Engineering; 2024. p. 1–13. -03-25].
- [23] Li Y, Zhang F, Li Y, et al. An improved two-stage robust optimization model for CCHP-P2G microgrid system considering multi-energy operation under wind power outputs uncertainties. *Energy* 2021;223:120048.
- [24] Zhang Y, Liu Y, Shu S, et al. A data-driven distributionally robust optimization model for multi-energy coupled system considering the temporal-spatial correlation and distribution uncertainty of renewable energy sources. *Energy* 2021; 216:119171.
- [25] Zhao B, Qian T, Tang W, et al. A data-enhanced distributionally robust optimization method for economic dispatch of integrated electricity and natural gas systems with wind uncertainty. *Energy* 2022;243:123113.
- [26] Yuan ZP, Xia J, Li P. Two-time-scale energy management for microgrids with data-based day-ahead distributionally robust chance-constrained scheduling. *IEEE Trans Smart Grid* 2021;(12–6).
- [27] Li Y, Han M, Shahidehpour M, et al. Data-driven distributionally robust scheduling of community integrated energy systems with uncertain renewable generations considering integrated demand response. *Appl Energy* 2023.
- [28] Dong Y, Zhang H, Ma P, et al. A hybrid robust-interval optimization approach for integrated energy systems planning under uncertainties. *Energy* 2023;274:127267.
- [29] Luan F, Choi J, Jung H. A particle swarm optimization algorithm with novel expected fitness evaluation for robust optimization problems. *IEEE Trans Magn* 2012;48(2):331–4.
- [30] Shen H, Zhang H, Xu Y, et al. Two stage robust economic dispatching of microgrid considering uncertainty of wind, solar and electricity load along with carbon emission predicted by neural network model. *Energy* 2024;300:131571.
- [31] Cai P, Yang M, Ma S, et al. Hierarchical game for integrated energy system and electricity-hydrogen hybrid charging station under distributionally robust optimization. *Energy* 2023;283:128471.
- [32] Hou J, Yu W, Xu Z, et al. Multi-time scale optimization scheduling of microgrid considering source and load uncertainty. *Elec Power Syst Res* 2023.
- [33] Tao H, Jincheng Z, Mohsen L. Stochastic energy scheduling in microgrid with real-time and day-ahead markets in the presence of renewable energy resources. *Soft Comput - A Fusion Found Methodol Appl* 2023;27(22):16881–96.
- [34] He Y, Li Z, Zhang J, et al. Day-ahead and intraday multi-time scale microgrid scheduling based on light robustness and MPC. *Int J Electr Power Energy Syst* 2023.
- [35] Jani A, Jadid S. Two-stage energy scheduling framework for multi-microgrid system in market environment. *Appl Energy* 2023;336:120683.
- [36] Xiao B, Zhang B, Wang X, et al. Short-term wind power interval prediction based on combined modal decomposition and deep learning. *Autom Electr Power Syst* 2023; 47(17):110–7.
- [37] Yang M, Guo Y, Huang Y. Wind power ultra-short-term prediction method based on NWP wind speed correction and double clustering division of transitional weather process. *Energy* 2023;282:128947.
- [38] Ge S, Fan Q, Liu H, et al. Distributionally robust optimization method for substation planning considering incentive response uncertainty. *Autom Electr Power Syst* 2023;47(24):80–9.
- [39] Liu Yixin, Guo Li, Wang Chengshan. Economic dispatch of microgrid based on two stage robust optimization. *Proceed. CSEE* 2018;38(14):4013–22. +4307.
- [40] Xiao H, Pu X, Pei W, et al. A novel energy management method for networked multi-energy microgrids based on improved DQN. *IEEE Trans Smart Grid* 2023;14 (6):4912–26.
- [41] Huang B, Wang J. Deep-reinforcement-learning-based capacity scheduling for PV-battery storage system. *IEEE Trans Smart Grid* 2021;12(3):2272–83.
- [42] Hu Junjie, Tong Yuxuan, Liu Xuetao, et al. Multi-time scale robust optimization strategy for comprehensive energy systems considering refined hydrogen utilization. *Trans China Electrotech Soc* 2024;39(5):1419–35.
- [43] Zhao T, Chen Q, Zhang J, et al. Energy/reserve co-scheduling of interconnected microgrids based on adaptive robust optimization. *Power Syst Technol* 2016;40 (12):3783–91.
- [44] Zhao M, Ding B, Wu Y, et al. Robust flexibility assessment of integrated energy systems based on conservative adaptability optimization[J/OL]. *Power Syst Technol* 2024;1–13. -08-06].
- [45] Tan M, Li Z, Su Y, et al. Dual time-scale robust optimization for energy management of distributed energy community considering source-load uncertainty. *Renew Energy* 2024;226:120435.